ESP-LRSMD: A Two-Step Detector for Ship Detection Using SLC SAR Imagery

Zongsen Lv, *Graduate Student Member, IEEE*, Jing Lu, Qing Wang, Zhengwei Guo, *Member, IEEE*, and Ning Li, *Member, IEEE*

***Abstract*— Synthetic aperture radar (SAR), an active microwave remote sensing equipment, can provide all-day, all-weather, and high-resolution images for Earth observation. Ship monitoring using SAR is significant to fishery application, marine transportation, maritime security, and so on. Most traditional SAR ship detection methods use the amplitude information of the SAR imagery and employ the constant false alarm rate (CFAR), deep learning, and other technologies to detect the ship targets. However, in addition to the amplitude image, the SAR single-look complex (SLC) data can also be utilized for ship detection since its complex-valued information is beneficial to characterize the ship's feature in the transforming domains. In this article, the strong scattering feature of ships in the image domain and low-rank feature in the transforming domain is studied, and on this basis, a two-step detector is proposed to detect ship targets in SAR SLC image. The first step is to identify the presence of ships, which is based on the eigen-subspace projection (ESP) technology to detect the occurrences of the ship targets in azimuth cells of the SAR data. The second step is to locate the position of ships, which is based on the low-rank and sparse matrix decomposition (LRSMD) to obtain the instantaneous frequency information of the ship targets in the time–frequency domain. Finally, the ship targets can be located in a line-by-line manner in the azimuth direction. Experimental results via European Sentinel-1A and Chinese Gaofen-3 single-pol SAR SLC data demonstrate the performance of the proposed detector.**

***Index Terms*— Eigen-subspace projection (ESP), low-rank and sparse matrix decomposition (LRSMD), ship detection, single-look complex (SLC) image, synthetic aperture radar (SAR).**

NOMENCLATURE

$\mathbf{S}$	Subspace matrix.
L	Dimension of subspace matrix.
N_{os}	Sampling number.
$\hat{\mathbf{R}}$	Covariance matrix.
$\mathbf{\Lambda}$	Matrix of eigenvalue.
$\mathbf{U}$	Matrix of eigenvector.
p	Sampling number in the time dimension.
q	Sampling number in the frequency dimension.
$\mathbf{D}$	Observation matrix.
$\mathbf{A}$	Low-rank matrix.
$\mathbf{E}$	Error matrix.
f_r	Range sampling frequency.
τ	Range time.
B	Signal bandwidth.
τ_0	Time delay.
$\mathbf{S_{tf}}$	STFT matrix of the signal $\zeta(\tau)$.
$\mathbf{X}$	Original matrix.
$\tilde{\mathbf{X}}$	Low-rank estimated matrix.
$\mathbf{T}_{\boldsymbol{n}}$	Subspace matrix of SAR pulse.
$\mathbf{\Omega}_n$	Covariance matrix of SAR pulse.
Φ	Maximum eigenvalue sequence of SAR data.
ξ_n	Maximum eigenvalue of SAR pulse.
α	Threshold.
$\mathbf{L}$	Instantaneous frequency components of the ship targets.
$\mathbf{C}$	Instantaneous frequency components of the sea clutters.
λ	Tuning parameter.
$\boldsymbol{\mathcal{Y}}$	Lagrange multiplier.
$\boldsymbol{\mathcal{U}}$	Left singular value vector.
$\boldsymbol{\mathcal{V}}$	Right singular value vector.
E	Iteration termination condition.
P^n	Position information of the ship targets.
N_τ	Total sampling points in time.
N_{win}	Length of time-limited window function.
N_{step}	Step length.
TP	True positives.
FP	False positive.
FN	False negatives.

Manuscript received 14 February 2022; revised 8 May 2022, 16 June 2022, and 12 July 2022; accepted 11 August 2022. Date of publication 16 August 2022; date of current version 30 August 2022. This work was supported in part by the National Natural Science Foundation of China under Grant 41901302 and Grant 61871175. *(Corresponding author: Ning Li.)*

Zongsen Lv, Zhengwei Guo, and Ning Li are with the Key Laboratory of Big Data Analysis and Processing of Henan Province, the Henan Engineering Research Center of Intelligent Technology and Application, and the College of Computer and Information Engineering, Henan University, Kaifeng 475004, China (e-mail: lvzongsen@henu.edu.cn; gzw@henu.edu.cn; hedalining@henu.edu.cn).

Jing Lu is with the Land Satellite Remote Sensing Application Center, Ministry of Natural Resources, Beijing 100048, China (e-mail: luj@lasac.cn).

Qing Wang is with the Air Force Research Institute, Beijing 100085, China (e-mail: wangqing_rs@pku.edu.cn).

Digital Object Identifier 10.1109/TGRS.2022.3198940

I. INTRODUCTION

A. Background

THE ocean area occupies 71.8% of the Earth's surface, which is the second space for humans to survive outside of land. The ship is a common means of water transportation, and it can take on tasks such as transportation, fishery, and national defense. Surveillance of ships is of great significance. With the development of remote sensing technology, it has become a hot spot to detect and locate maritime ship targets through remote sensing images [1].

Synthetic aperture radar (SAR), an active microwave remote sensing equipment that can dynamically observe the Earth all-day and all-weather, has significant potential in remote sensing [2], [3]. The resolution of SAR does not depend on

1558-0644 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:23:47 UTC from IEEE Xplore. Restrictions apply.

the observation distance because it uses pulse compression technology to obtain a high resolution of the target in range and uses the synthetic aperture technology to obtain the high resolution of the target in azimuth. In addition, spaceborne SAR satellites can observe the ocean circularly in a certain revisit period to obtain time-series SAR images, which is beneficial to ocean remote sensing. Therefore, using SAR to monitor ships is appropriate compared with infrared and optical passive remote sensing equipment, and it is beneficial for sustainable fishing, marine ecosystem protection, and military target strikes. However, influenced by speckle noise, clutter, interference, and other factors [4], it is challenging to detect and locate ships in SAR images.

B. State of the Art

In the past few decades, numerous SAR ship detection methods have been researched, which can be roughly divided into four categories, namely, statistical model-based methods, deep learning (DL)-based methods, superpixel-based methods, and complex information-based methods [5], [6], [7].

1) Statistical Model-Based Methods: For the statistical model-based methods, the constant false alarm rates (CFARs) and its modifications are best known, which have been widely used in the ship target detection system. Generally, the ships are made of metal, which means that the scattering intensities of the ship targets are higher than sea clutter. Inspired by this prior information, CFAR statistically models the sea clutter, solves the false alarm probability threshold, and uses threshold segmentation to realize ship target detection [8]. Different statistical distributions to fit the heterogeneous sea clutter have been researched to boost the detection performance, including Weibull distribution [9], K distribution [10], [11], compound Gaussian distribution [12], generalized Gamma distribution [13], Gamma distribution [14], Rayleigh [15], and 2-D joint log-normal [16], [17] distribution, and three-parameter generalized-K (GK) [18], [19]. Besides, while researching the model of sea clutter distribution, in order to obtain an accurate threshold, the design of modified CFAR detector has also attracted the attention of scholars. Methods, including cell-averaging CFAR [20], greatest of CFAR [21], and wake CFAR [22], are proposed to detect ships under inhomogeneous sea clutter.

From an electromagnetic point of view, a ship can be considered as a dominant scatterer characterized by a strong and coherent backscattered signal, and therefore, a series of physical-model-based approaches is proposed. A previous study in [23] proposed an effective filtering technique, which is based on the use of the Rice factor parameter to highlight the presence of the ship targets. In [24], a CFAR filter based on the coherent-to-incoherent field ratio is developed. This method is characterized by a very low false alarm rate and can operate in a wide range of incidence angles.

2) DL-Based Methods: Recently, with the rapid development of DL theory, it has been successfully applied in the fields of computer vision and pattern recognition [25]. The robust feature representation ability of the DL technology has caused scholars in the SAR ship detection community to start to pay much attention. Many publicly available datasets, including SAR ship detection dataset (SSDD) [26], Gaofen-3 ship detection dataset (GF3SDD) [27], and SAR ship segmentation dataset (SAR-HR4) [28], are established to research the DL-based SAR ship detection method. A series of state-of-the-art models is proposed to detect ship targets based on the traditional deep neural network framework, i.e., convolutional neural network (CNN). For instance, fast regions with CNN (R-CNN) and its modifications are designed for extracting the shiplike region of interests (ROIs) [29]. A method based on faster R-CNN is proposed to improve the detection performance and speed up the training time [30]. Besides, a dense pyramid network based on a pyramid structure [31] and part-based CNN [32] is proposed to detect the ship targets in complex scenes. The above studies can be regarded as two-stage detection methods, and they usually have high detection performance but are computationally prohibitive. In contrast to the two-stage methods, one-stage methods, including you only look once-based method [33] and its modifications, single-shot multibox detector [34], and RetinaNet [35], show faster running speed. It is noteworthy that the grid CNN-based method proposed in [36] can be applied to detect ship targets in real time. Many of the target detection networks employ the intersection over union (IoU) to detect targets, which has great limitations for some small ship target detection. Recently, the method in [37] employs the Bhattacharyya distance instead of the IoU metric to improve the performance of small ship target detection.

3) Superpixel-Based Methods: The state-of-the-art superpixel technology is a hot topic in the field of image segmentation. In [38], a method combining superpixel with CFAR technology, i.e., the superpixel level CFAR (SP-CFAR), is proposed. The SP-CFAR employs superpixels technology instead of slide windows operation of CFAR, and the noise (clutter and interference) pixels can be removed. In [39], based on the superpixel technology, the Bhattacharyya distance is considered to distinguish the ship targets from the sea clutter. The method in [40] considers using the Fisher vector (FV) as an alternative to superpixels for ship detection. The FV uses the gradient vector of the log-likelihood function to express an image. Thus, superpixels' zero-, first-, and second-order characteristics can be captured simultaneously, and a better detection performance can be obtained. In addition, in order to further improve the detection performance, labeled FVs-based detector and improved FV-based detector are proposed [41]. Recently, the method in [42] employs a saliency analysis of multiscale superpixels to detect the ship targets on SAR imagery. In contrast to the traditional superpixel-based methods, the proposed method dynamically generates superpixels, which improves the detection accuracy. In addition, a method [43] integrating superpixel generation and an optimized CFAR technology is proposed. This method avoids the drawbacks of sliding window in the traditional CFAR and saves unnecessary time consumption.

4) Complex Information-Based Methods: Most of the existing SAR ship detection methods are based on the amplitude information of the image because scholars believe that the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:23:47 UTC from IEEE Xplore. Restrictions apply.

phase information of the image is randomly and nonuniformly distributed [44]. Recently, related studies have shown that phase or complex information is beneficial for distinguishing ships and sea clutter in high-resolution SAR images [45]. In [46], a complex domain statistical model of high-resolution SAR image is built. Subsequently, the method in [47] is based on the statistical model and employs the complex signal kurtosis (CSK) to detect ships in complex-valued SAR imagery. Since the CSK-based method uses the complex information of the SAR image, the detection results avoid the high false alarm rate caused by interference, azimuth ambiguity, and other factors.

C. Problem Statement

Most existing methods are based on statistical modeling and feature representation technology to locate and detect ships on SAR amplitude images. Methods based on statistical modeling need to establish a statistical model of sea clutter accurately; otherwise, the detection performance will be affected. The DL-based methods are supervised, which means that they need to train a large number of training samples and to continuously fit the model through training. The DL-based methods have limitations in the two aspects. First, the framework of the model needs robust generalization ability to realize ship detection under different scenes, sea levels, and angles of view. Second, a large number of training processes are computationally complex, which means that it is challenging to detect ships in real time.

Besides, most SAR ship detection methods have the disposition to employ amplitude information, whereas the potential of the complex information could be underrated. The complex-valued information of the SAR single-look complex (SLC) image, which is beneficial to characterize the ship's feature in the transforming domains, is not being fully exploited so far. The few existing complex information-based methods rely on statistical modeling, and they establish a complex domain statistical model of the SAR imagery. Nevertheless, the model of the sea clutter is often difficult to establish accurately. These limitations bring challenges to the detection task.

D. Solution and Contributions

In this article, a novel detector, which is a two-step process integrating eigen-subspace projection (ESP) with low-rank and sparse matrix decomposition (LRSMD), namely, ESP-LRSMD, is proposed to detect ship targets in the SAR SLC image. The first step in the proposal is to detect the occurrences of the ship targets in azimuth cells of the SAR SLC data, which is done in the eigen-subspace domain. Within the step, the Hankel structure and eigenvalue decomposition (EVD) is employed to project the SAR SLC data into eigen-subspace. Then, the occurrences of the ship targets can be detected. The second step is to locate the ship's position, which is done in the time–frequency domain. Within the step, the short-time Fourier transform (STFT) transforms the SAR SLC data into the time–frequency domain. Then, the LRSMD technology is employed to decompose the time–frequency matrix into a low-rank matrix (i.e., the instantaneous frequency component of the ship targets) and a sparse matrix (i.e., the instantaneous frequency component of the sea clutter). Finally, by accumulating the energy of the STFT spectrum, the ship targets can be detected in a line-by-line manner in the azimuth direction.

The main contributions of this article are summarized as follows.

1) Two characteristics of the ship target, i.e., the strong scattering feature in the eigen-subspace domain and the low-rank feature in time–frequency domain, are researched. Based on the theoretical foundation and experimental analysis, these characteristics are analyzed and verified.
2) A novel detector, which is a two-step process integrating ESP with LRSMD technology, is proposed to detect ship targets in the SAR SLC image. Compared with the traditional detectors, the proposal is not model-driven, and it makes full use of the information of complex-valued SAR imagery, which can detect ship targets under different sea states accurately.

E. Organization

The remainder of this article is organized as follows. In Section II, the basic theories of the related works are briefly introduced. In Section III, the characteristics of ship targets in the time domain and time–frequency domain are researched. In Section IV, the proposed two-step detector for ship targets detection is presented in detail. In Section V, experimental results and the performance analysis of the proposal are given, and finally, Section VI concludes this article.

F. Notations

The notations in this article are explained in the Nomenclature.

II. Related Works

A. Eigen-Subspace Projection

The ESP technology is common in sparse signal estimation, direction of arrival, and other fields [48], [49], [50]. It decomposes the autocorrelation matrix of observed signal by EVD. According to the distribution of eigenvalues, the subspace of the signal of interest (SOI) can be constructed. By using the orthogonality between the subspaces, the SOI can be projected.

Denoting the dimension of subspace matrix by L, $M = N_{\text{os}} + 1 - L$, where N_{os} is the sampling number. By using the Hankel structure, the subspace matrix $\mathbf{S} \in \mathbb{C}^{L\times M}$ of the observed signal $s(p) \in \mathbb{C}^{1\times N_{\text{os}}}$ can be constructed as follows:

$$\mathbf{S} = \begin{bmatrix} s(1) & s(2) & s(3) & \cdots & s(M) \\ s(2) & s(3) & s(4) & \cdots & s(M+1) \\ s(3) & s(4) & s(5) & \cdots & s(M+2) \\ \vdots & \vdots & \vdots & \ddots & \cdots \\ s(L) & s(L+1) & s(L+2) & \cdots & s(M+L-1) \end{bmatrix}. \tag{1}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:23:47 UTC from IEEE Xplore. Restrictions apply.

The covariance matrix $\hat{\mathbf{R}} \in \mathbb{C}^{L\times L}$, which is a symmetric and positive semidefinite matrix, can be constructed from matrix $\mathbf{S}$

$$\hat{\mathbf{R}} = \mathbf{S}\mathbf{S}^{\mathrm{H}}. \tag{2}$$

Then, by the EVD, the covariance matrix $\hat{\mathbf{R}}$ can be decomposed

$$\hat{\mathbf{R}} = \mathbf{S}\mathbf{S}^{\mathrm{H}} = \mathbf{U}\Lambda\mathbf{U}^{\mathrm{H}} \tag{3}$$

where $\mathbf{\Lambda} = \mathrm{diag}(\lambda_1, \lambda_2, \lambda_3, \ldots, \lambda_L)$, $\mathbf{\Lambda}$ is an $L \times L$ diagonal complex matrix, and λ is the eigenvalue and the corresponding eigenvector is $\mathbf{U} = [u_1, u_2, u_3, \ldots, u_L]$. The subspace constructed by the r dominant eigenvectors is called the SOI subspace, which can be expressed as

$$\mathbf{U_r} = \mathrm{span}\{u_1, u_2, u_3, \ldots, u_r\}, \quad 1 \le r \le L. \tag{4}$$

$\mathbf{S}$ dcan be projected into the SOI subspace by the following function:

$$\mathbf{S_j} = \mathbf{U_r}\mathbf{U_r^H}\mathbf{S}. \tag{5}$$

Finally, the SOI can be obtained by the following:

$$s_{\mathrm{SOI}}(p) = \begin{cases} \frac{1}{2}\left[\mathbf{S_j}(1, p) + \mathbf{S_j}(p, 1)\right], & 1 \le p \le M-1 \\ \frac{1}{2}\left[\mathbf{S_j}(p-M+1, M) + \mathbf{S_j}(p, 1)\right], & M \le p \le L \\ \frac{1}{2}\left[\mathbf{S_j}(p-M+1, M) + \mathbf{S_j}(L, p-L+1)\right], & L+1 \le p \le N_{\mathrm{os}}. \end{cases} \tag{6}$$

B. Short-Time Fourier Transform

The Fourier transform (FT) can characterize the signal in the frequency domain but cannot describe the characteristics of the signal changing with time. In 1946, Gabor proposed the STFT [51]. As a fundamental time–frequency analysis method, the STFT can obtain time and frequency characters as well as their instantaneous relationship for a signal.

The main idea of STFT is that a time-limited window function $h(p)$ is employed to multiply the signal before the processing of FT, meanwhile, assuming that the signal is stable (pseudostationary) within a short time interval of the analysis window. Moving the window function so as to analyze the signal in a section-by-section manner obtains a set of local spectrum of the signal. The discrete STFT spectrum of signal $s(p)$ can be expressed as

$$\mathrm{STFT}(p, q) = \sum_u h(u-p)s(u)\exp\left(-j\frac{2\pi}{N_r}qu\right) \tag{7}$$

where p and q denote the sampling number in time and frequency, respectively, and u is the signal integer index.

C. Low-Rank and Sparse Matrix Decomposition

In the era of big data, large-scale data often contain much redundant information. Mathematically, the matrix form of these data is low-rank (or approximate low-rank). The low-rank component of the observed matrix plays the analogous role of sparsity in the field of compressive sensing, and it can be regarded as a dominant low-dimensional description of the data. However, the sparsity error under Gaussian distribution will destroy the low-rank feature of the data. The low-rank and sparse model was built to restore the low-rank characteristics of the matrix, and scholars usually employ optimization methods to solve problems [51], [52].

Consider the observation matrix $\mathbf{D}$ with low-rank component $\mathbf{A}$ and unknown error component $\mathbf{E}$, and the low-rank characteristics can be restored by solving the following problem, which can be expressed as:

$$\min_{A,E} \|\mathbf{E}\|_F \quad \text{s.t. } \mathrm{rank}(\mathbf{A}) \le r, \quad \mathbf{D} = \mathbf{A} + \mathbf{E} \tag{8}$$

where $\mathrm{rank}(\cdot)$ is the rank operation. $\|\mathbf{E}\|_F$ is the Frobenius norm of matrix $\mathbf{X}$, which is defined as

$$\|\mathbf{E}\|_F = \sqrt{\sum_i \sum_j |a_{i,j}|^2}. \tag{9}$$

The LRSMD seeks to decompose a matrix into a sum of two components: one has a low-rank feature and the other is being sparse. In case the matrix $\mathbf{E}$ is sparse, (8) can be transformed into an optimization problem of double-objective functions

$$\min_{A,E} (\mathrm{rank}(\mathbf{A}), \|\mathbf{E}\|_0) \quad \text{s.t. } \mathbf{D} = \mathbf{A} + \mathbf{E} \tag{10}$$

where $\|\mathbf{E}\|_0$ denotes the l^0 norm of $\mathbf{E}$.

Then, by introducing a tuning parameter, (10) can be transformed into an optimization problem of single-objective functions

$$\min_{A,E} \mathrm{rank}(\mathbf{A}) + \lambda\|\mathbf{E}\|_0 \quad \text{s.t. } \mathbf{D} = \mathbf{A} + \mathbf{E} \tag{11}$$

where λ is the tuning parameter. Since the rank operation and l^0 norm are nonconvex and nonsmoothness, the minimization problem (11) is NP-hard. Therefore, the shrink operation is needed to further improve the problem. Candes *et al.*, who provided a solution [52], transformed the NP-hard problem (11) into a convex approximation

$$\min_{A,E} \|\mathbf{A}\|_* + \lambda\|\mathbf{E}\|_1 \quad \text{s.t. } \mathbf{D} = \mathbf{A} + \mathbf{E} \tag{12}$$

where the first term of (12) is the nuclear norm of $\mathbf{A}$ and it takes the place of the rank operation, whereas the second term of (12) is the ℓ^1 norm of $\mathbf{E}$ and it takes the place of the ℓ^0 norm. The nuclear norm is the sum of singular values of the matrix. Problem (12) is also named the robust principal component analysis (RPCA) problem.

III. On the Characteristics of the Ship Target in the Transforming Domains

A. On the Strong Scattering Intensity of the Ship Targets in Time Domain

In general, the ships are made of metal, and their scattering intensities are higher than sea clutter. In the background of sea

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:23:47 UTC from IEEE Xplore. Restrictions apply.

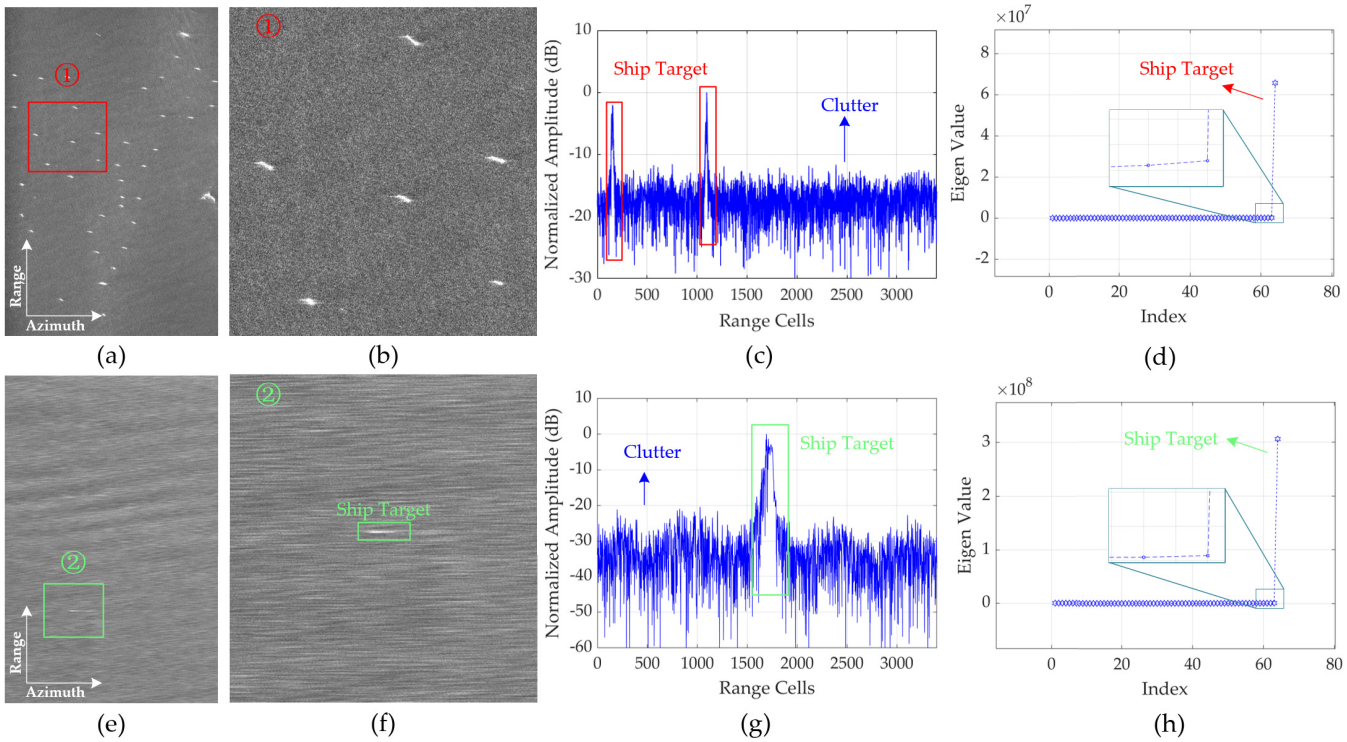


Fig. 1. Strong scattering feature of the ship targets in the time domain and eigen-subspace. (a) Amplitude image of Sentinel-1A. (b) Enlarged area in (a). (c) Pulse contains two ship targets in (b). (d) Eigenvalue of the signal in (c). (e) Amplitude image of GF-3. (f) Enlarged area in (e). (g) Pulse contains one ship target in (f). (h) Eigenvalue of the signal in (g).

clutter, the ship targets are characterized by strong scattering and sparse distribution. Based on this feature, traditional detectors distinguish between ship targets and sea clutter by constructing the sea clutter model or DL model. In this article, based on the ESP technology, the authors project the signal containing ship targets into the eigen-subspace and find that the strong scattering feature of the ship targets can also be characterized. Interestingly, the energy of the ship target constitutes the principal component of the signal, which is because the mapping of the eigen-subspace is linear, and its transformation does not change the characteristics of ship targets in the time domain.

To further characterize the ship targets in the time domain, two example cases, i.e., a scene under normal sea states observed by European Sentinel-1A satellite working on stripmap mode and a scene under severe sea states observed by Chinese Gaofen-3 (GF-3) satellite working on sliding spotlight mode, are provided and analyzed, which is shown in Fig. 1. Fig. 1(a) and (b) shows the amplitude image of Sentinel-1A and its enlarged area, respectively. Within the figures, ships are scattered on the sea and visible. Fig. 1(c) shows an SOI (azimuth cell) containing ship targets, where the spikes are the ship targets. Compared with clutter, the scattering intensity of ships is significantly high. Fig. 1(d) shows the eigenvalue representation by the ESP technology of the SOI in Fig. 1(c), where the dominant eigenvalues represent the principal component (two ship targets) of the SOI. Fig. 1(e) and (f) shows the amplitude image of Gaofen-3 and its enlarged area, respectively. Within the figures, the sea states are complex and severe. In such a complex scene, the sea surface is inhomogeneous influenced by the ocean wind, whereas the scattering intensity of the ship target is still higher than that of sea clutter. The strong scattering feature can be seen in the SOI of Fig. 1(f) and its eigen-subspace, which is shown in Fig. 1(g) and (h).

B. On the Low-Rank of the Ship Targets in Time–Frequency Domain

The ship targets in SAR SLC data are low-rank in the time–frequency domain. Generally, SAR transmits linear frequency modulation (LFM) signal with a high time–bandwidth product and employs pulse compression technology to obtain a high resolution of the target in range. Within SAR imaging, matched filtering (MF), a way to pulse compression, can eliminate the secondary phase of the signal, and as a result, a flat spectrum with a linear phase can be obtained, which can be expressed as

$$S_{\text{out}}(f_r) = \text{rect}\left(\frac{f_r}{B}\right)\exp\{-j2\pi f_r \tau_0\} \tag{13}$$

where f_r, B, and τ_0 denote the range sampling frequency, signal bandwidth, and time delay, respectively.

By employing the FT, the signal can be focused as a point target (similar to the sinc function), which can be expressed as

$$s_{\text{out}}(\tau) = B\text{sinc}\{B(\tau - \tau_0)\} \tag{14}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:23:47 UTC from IEEE Xplore. Restrictions apply.

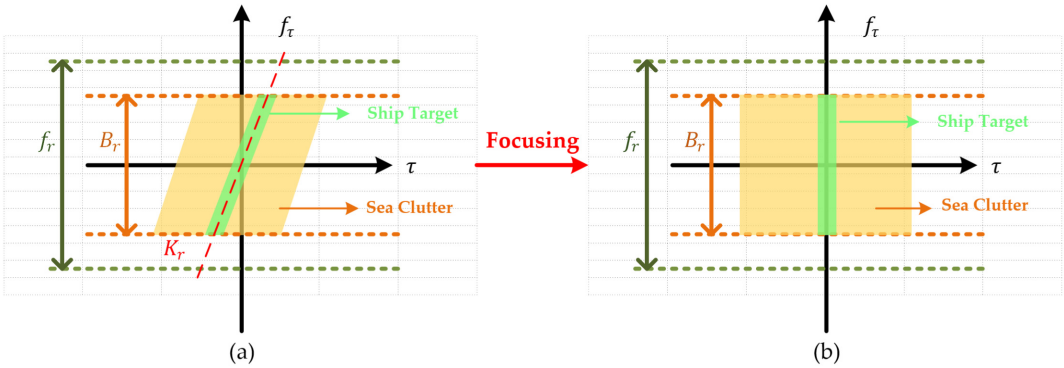


Fig. 2. Time–frequency diagram of SAR data before and after imaging process. (a) Time–frequency history of the SAR raw data. (b) Time–frequency history of the SAR SLC image.

where τ denotes the range time. The time–frequency diagram of SAR data before and after imaging process is shown in Fig. 2. For the real ocean scene of the SAR imagery containing ship targets, sea clutter, and so on, denoting the signal in azimuth cells that contains ship targets and sea clutter by $\zeta(\tau)$, the STFT is employed to characterize the time–frequency characteristics, which can be expressed as

$$\mathrm{STFT}[\zeta(\tau)] = \mathbf{S_{tf}} = \begin{bmatrix} \mathcal{C}_1 \mathrm{rect}(f_r/B)\exp\{-j2\pi f_r \tau_1\} \\ \mathcal{C}_2 \mathrm{rect}(f_r/B)\exp\{-j2\pi f_r \tau_2\} \\ \vdots \\ \mathcal{C}_{N_p} \mathrm{rect}(f_r/B)\exp\{-j2\pi f_r \tau_{N_p}\} \end{bmatrix}^T \tag{15}$$

where $\mathbf{S_{tf}}$ denotes the STFT matrix of the signal $\zeta(\tau)$ and $\tau_1, \tau_2, \ldots, \tau_{N_p}$ denotes the local time caused by the time-limited window function $h(p)$. Each row represents the instantaneous spectrum in local time, and $\mathcal{C}_1, \mathcal{C}_2, \ldots, \mathcal{C}_{N_p}$ denote the magnitude of the instantaneous spectrum.

As aforementioned, the scattering intensities of the ship targets are higher than the sea clutter. The power of the instantaneous spectrum amplitude of the ship targets is also higher than that of the sea clutter. Within formula (15), each line's rectangular function and linear phase term are approximately equal, whereas the variant is the amplitude of the instantaneous spectrum. Under the assumption that there exist N_{ship} instantaneous frequency components of the ship targets and the instantaneous frequency of sea clutter is small so that it can be ignored, formula (15) can be approximated as

$$\mathbf{S_{tf}} \approx \tilde{S}_{\mathbf{tf}} = \begin{bmatrix} \mathcal{C}_1 \mathrm{rect}\left(\frac{f_r}{B}\right)\exp\{-j2\pi f_r \tau_1\} \\ \mathcal{C}_2 \mathrm{rect}\left(\frac{f_r}{B}\right)\exp\{-j2\pi f_r \tau_2\} \\ \vdots \\ \mathcal{C}_{N_{\text{ship}}} \mathrm{rect}\left(\frac{f_r}{B}\right)\exp\{-j2\pi f_r \tau_{N_{\text{ship}}}\} \\ \mathbf{0} \\ \vdots \\ \mathbf{0} \end{bmatrix}^T \tag{16}$$

where $\tilde{S}_{\mathbf{tf}}$ denotes a low-dimensional description of STFT matrix $\mathbf{S_{tf}}$ and $\tilde{S}_{\mathbf{tf}}$ is low-rank and $\mathrm{rank}(\tilde{S}_{\mathbf{tf}}) = N_{\text{ship}}$.

The SOI shown in Fig. 1(c) that contains ship targets is employed to analyze the low-rank feature of the ship targets in the time–frequency domain, which is shown in Fig. 3. In Fig. 3(a), the frequency components of the ship target show linear characteristics and are marked by the red rectangle. Fig. 3(b) and (c) shows the top 300 in 1024 singular values and low-rank estimation errors (LREEs), where the LREE is defined as

$$\mathrm{LREE}(\mathbf{X}, \tilde{\mathbf{X}}) = \frac{\|\mathbf{X} - \tilde{\mathbf{X}}\|_F^2}{\|\mathbf{X}\|_F^2} \tag{17}$$

where $\mathbf{X}$ and $\tilde{\mathbf{X}}$ denote the original matrix and the estimated matrix, respectively.

As shown in Fig. 3(b), the top 10 singular values are significantly larger than others, and they represent the singular value of the ship targets. The estimation errors under different rank approximations are shown in Fig. 3(c). It can be seen that lower rank can reconstruct a low-rank approximation matrix with a smaller estimation error. Fig. 3(d)–(f) shows the rank-1, rank-3, and rank-5 approximation matrix, respectively, and Fig. 4(g)–(i) shows their residual matrices. Within the figures, the low-rank approximation matrix characterizes the instantaneous frequency component of the ship target, whereas the residual matrix characterizes the sea clutter. It is worth noting that when using rank-1 approximation, LREE can reach 0.1692, which means that rank-1 approximation can characterize the principal components of the time–frequency matrix, i.e., the rank-1 approximation can estimate the instantaneous frequency component of the ship target.

As will be discussed later, the strong scattering intensity and the low-rank feature form the basis of this article's proposal.

IV. Methodology: A Two-Step Detector for SAR Ship Detection

In this article, by utilizing the strong scattering feature of ships in the time domain and low-rank feature in the time–frequency domain, a two-step detector, which is based on the ESP and LRSMD technology, is proposed. The first step is

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:23:47 UTC from IEEE Xplore. Restrictions apply.

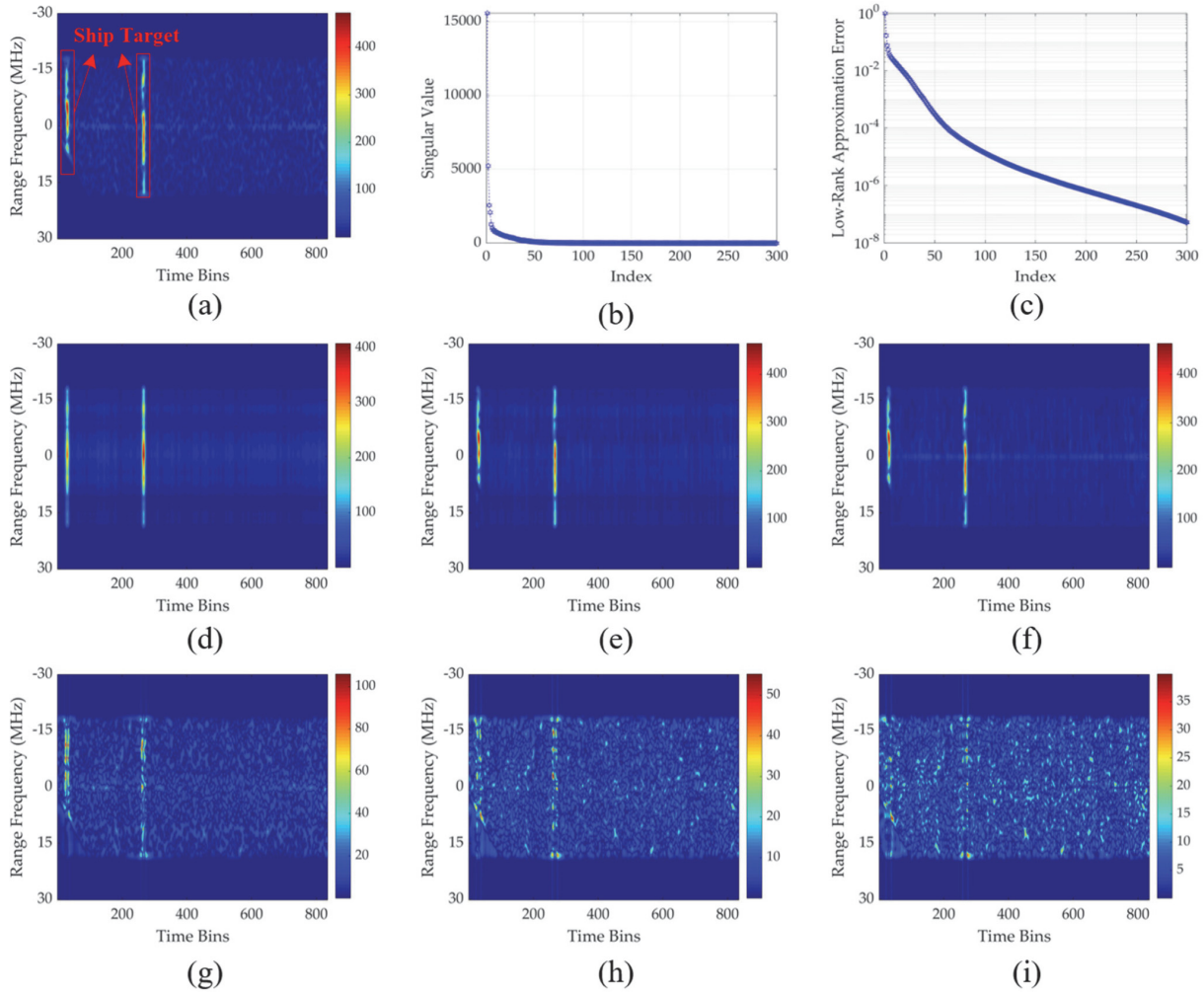


Fig. 3. Time–frequency spectrum and singular values of SOI in Fig. 1(c). (a) Time–frequency spectrum of SOI in Fig. 1(c). (b) Top 300 in 1024 singular values. (c) LREE for rank approximation. (d) Rank-1 approximation matrix, LREE = 0.1692. (e) Rank-3 approximation matrix, LREE = 0.0528. (f) Rank-5 approximation matrix, LREE = 0.0325. (g)–(i) Residual matrix of (d)–(f), respectively.

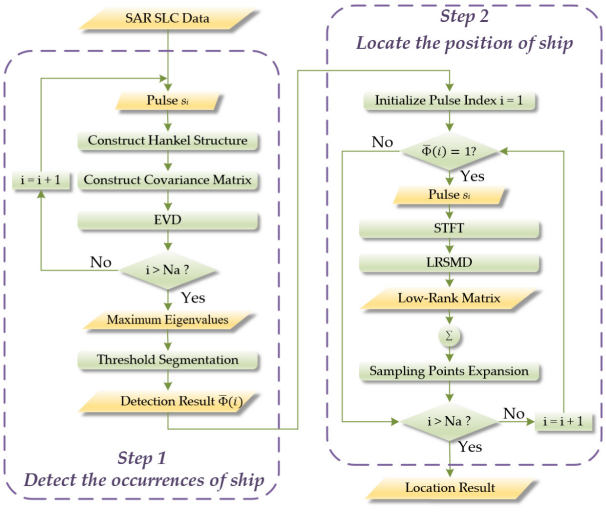


Fig. 4. Flowchart of the proposed detector.

to identify the presence of ships, whereas the second step is to locate the position of ships. The flowchart of the proposed detector is shown in Fig. 4.

A. Identify the Presence of Ships

1) Obtain the Maximum Eigenvalue: The traditional SAR ship detection methods are based on a primary hypothesis to process SAR images, i.e., the ship targets occur more or less in SAR imagery. However, for the real-time detection task, the occurrence of ships is an uncertain factor, and blind detection and processing undoubtedly increase the complexity of the task. In this article, the proposed detector preliminarily detects the occurrence of ships in azimuth cells, which is beneficial to the subsequent positioning of the ship.

Denote the SAR SLC data by $\mathbf{S}_{st} \in \mathbb{C}^{N_a \times N_r}$, where $\mathbf{S}_{st} = [s_{\mathrm{st}}^1, s_{\mathrm{st}}^2, \ldots, s_{\mathrm{st}}^{Na}]^T$ and $s_{\mathrm{st}}^n(\tau) \in \mathbb{C}^{1\times N_r}, \tau = 1, 2, \ldots, N_r$. By using the Hankel structure, the subspace matrix of $s_{\mathrm{st}}^n(\tau)$ can be expressed as

$$\mathbf{T}_n = \begin{bmatrix} s_{\mathrm{st}}^n(1) & s_{\mathrm{st}}^n(2) & s_{\mathrm{st}}^n(3) & \cdots & s_{\mathrm{st}}^n(M) \\ s_{\mathrm{st}}^n(2) & s_{\mathrm{st}}^n(3) & s_{\mathrm{st}}^n(4) & \cdots & s_{\mathrm{st}}^n(M+1) \\ s_{\mathrm{st}}^n(3) & s_{\mathrm{st}}^n(4) & s_{\mathrm{st}}^n(5) & \cdots & s_{\mathrm{st}}^n(M+2) \\ \vdots & \vdots & \vdots & \ddots & \cdots \\ s_{\mathrm{st}}^n(L) & s_{\mathrm{st}}^n(L+1) & s_{\mathrm{st}}^n(L+2) & \cdots & s_{\mathrm{st}}^n(M+L-1) \end{bmatrix} \tag{18}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:23:47 UTC from IEEE Xplore. Restrictions apply.

where $M = N_r + 1 - L$ and $n = 1, 2, \ldots, N_a$. The setting of L (i.e., the dimension of the subspace matrix) affects the accuracy of detection. Specifically, in case L is too small, the decomposition is not sufficient, which degrades the resolution of the ship in subspace; in case L is too large, the amount of calculation increases and the detection accuracy will not be improved. The size of L is set according to the sea-state level and the focus of the ship, and usually, 32 or 64 is considered [50].

According to formula (2), the covariance matrix can be expressed as

$$\mathbf{\Omega}_n = \mathbf{T}_n \mathbf{T}_n^H. \tag{19}$$

Then, perform the processing of EVD

$$\mathbf{\Omega}_n = \mathbf{T}_n \mathbf{T}_n^H = \mathbf{E}_n \mathbf{\Lambda}_n \mathbf{E}_n^H \tag{20}$$

where $\mathbf{\Lambda}_n = \text{diag}(\gamma_1^n, \gamma_2^n, \gamma_3^n, \ldots, \gamma_L^n)$ and γ^n is the eigenvalue and the corresponding eigenvector is $\mathbf{E} = \left[\epsilon_1^n, \epsilon_2^n, \epsilon_3^n, \ldots, \epsilon_L^n\right]$. Repeating the above process in a line-by-line manner and obtaining the maximum eigenvalue after each processing of EVD, the maximum eigenvalue sequence can be obtained, which can be expressed as

$$\Phi(n) = \left[\xi_1, \xi_2, \ldots, \xi_{N_a}\right] \tag{21}$$

where $\xi_n = \max[\gamma_1^n, \gamma_2^n, \gamma_3^n, \ldots, \gamma_L^n]$.

B. Threshold Segmentation

Within the maximum eigenvalue sequence $\Phi(n)$, according to the strong scattering intensity of ships in the time domain, it can be known that there exists a significant difference between the maximum eigenvalues of the azimuth cells containing the ship target and maximum eigenvalues of other azimuth cells. The robust K-means algorithm [53] is employed to divide $\Phi(i)$ into two categories, and one of them indicates the presence of ships, while the other indicates the clutters. The operation of threshold segmentation can be expressed as

$$\bar{\Phi}(n) = \begin{cases} 0, & \Phi(n) < \alpha \\ 1, & \Phi(n) \geq \alpha \end{cases} \tag{22}$$

where $\bar{\Phi}(n)$ is the detection result, α is the threshold value, and constants 1 and 0 label the occurrence of the ships and clutters, respectively.

C. Locate the Position of Ships

By transforming the SAR image to the time–frequency domain and utilizing the low-rank feature of the ship target and the sparse feature of the sea clutter, the ship detection problem can be regarded as a matrix decomposition problem, i.e., seek a feasible scheme to decompose the time–frequency matrix into a low-rank one and a sparse one. In this article, inspired by the LRSMD problem, a model is built for extracting the low-rank matrix from the time–frequency matrix, which can be expressed as

$$\text{STFT}\left[s_{\text{st}}^n(\tau)\right] = \mathbf{M} = \mathbf{L} + \mathbf{C} \tag{23}$$

where $\mathbf{L} \in \mathbb{C}^{N_f \times N_t}$ is the low-rank matrix and it denotes the instantaneous frequency components of the ship targets; $\mathbf{C} \in \mathbb{C}^{N_f \times N_t}$ is the sparse matrix and it denotes the instantaneous frequency components of the sea clutters and obeys independent identically distributed; and N_f and N_t denote the sampling point in frequency and time, respectively.

1) Obtain the Low-Rank Matrix: The model for extracting the low-rank matrix from the time–frequency matrix can be expressed as

$$\begin{aligned} &\min_{\mathbf{L},\mathbf{C}} \|\mathbf{L}\|_* + \lambda \|\mathbf{C}\|_1 \\ &\text{s.t. } \mathbf{M} = \mathbf{L} + \mathbf{C} \end{aligned} \tag{24}$$

where λ denotes the tuning parameter and it can be set as

$$\lambda = \frac{1}{\sqrt{\max(N_f, N_t)}}. \tag{25}$$

The augmented Lagrange multiplier method is employed to solve the problem. Problem (24) can be transformed into an unconstrained optimization problem, which can be expressed as

$$\min_{\mathbf{L},\mathbf{C}} \mu \|\mathbf{L}\|_* + \lambda\mu \|\mathbf{C}\|_1 + \frac{1}{2}\|\mathbf{M} - \mathbf{L} - \mathbf{C}\|_F^2 \tag{26}$$

where $\mu = (\sqrt{N}_f + \sqrt{N}_t)$. Then, the augmented Lagrange function of (26) can be expressed as

$$\min_{\boldsymbol{\mathcal{X}},\boldsymbol{\mathcal{Z}}} f(\boldsymbol{\mathcal{X}}) + g(\boldsymbol{\mathcal{Z}}) + \langle \boldsymbol{\mathcal{Y}}, \boldsymbol{\mathcal{X}} - \boldsymbol{\mathcal{Z}} \rangle + \frac{\rho}{2}\|\mathbf{M} - \mathbf{L} - \mathbf{C}\|_F^2 \tag{27}$$

where $f(\boldsymbol{\mathcal{X}}) = \mu(\|\mathbf{L}_{\boldsymbol{\mathcal{X}}}\|_* + \lambda\|\mathbf{C}_{\boldsymbol{\mathcal{X}}}\|_1)$ and $\boldsymbol{\mathcal{X}} = [\mathbf{L}_{\mathcal{X}}, \mathbf{C}_{\mathcal{X}}]$; $g(\boldsymbol{\mathcal{Z}}) = (1/2)\|\mathbf{M} - \mathbf{L}_{\boldsymbol{\mathcal{Z}}} - \mathbf{C}_{\boldsymbol{\mathcal{Z}}}\|_F^2$ and $\boldsymbol{\mathcal{Z}}\boldsymbol{\mathcal{Y}} = [\mathbf{L}_{\mathcal{Z}}, \mathbf{C}_{\mathcal{Z}}]$; $\boldsymbol{\mathcal{Y}}$ denotes the Lagrange multiplier and $\boldsymbol{\mathcal{Y}} = [\mathbf{L}_{\mathcal{Y}}, \mathbf{C}_{\mathcal{Y}}]$; and ρ is the relaxation parameter.

Obtain the optimal solution of $\boldsymbol{\mathcal{X}}$ and $\boldsymbol{\mathcal{Z}}$ in an alternating manner. In the $(k+1)$th iteration, fix $\boldsymbol{\mathcal{Z}}$ and $\boldsymbol{\mathcal{Y}}$, and solve the optimal solution of $\boldsymbol{\mathcal{X}}$

$$\boldsymbol{\mathcal{X}}_{k+1} = \min_{\boldsymbol{\mathcal{X}}} f(\boldsymbol{\mathcal{X}}) + \langle \boldsymbol{\mathcal{Y}}_{\boldsymbol{k}}, \boldsymbol{\mathcal{X}} \rangle + \frac{\rho}{2}\|\mathbf{M} - \mathbf{C}_{\mathcal{X}}\|_F^2. \tag{28}$$

Formula (28) can be equivalent to

$$\begin{bmatrix} \mathbf{L}_{\mathcal{X}_{k+1}} \\ \mathbf{C}_{\mathcal{X}_{k+1}} \end{bmatrix} = \begin{bmatrix} \min\limits_{L_{\boldsymbol{\mathcal{X}}}} \left(\mu\|\mathbf{L}_{\mathcal{X}}\|_* + \frac{\rho}{2}\left\|\mathbf{L}_{\mathcal{X}} - \mathbf{L}_{\mathcal{Z}_k} + \frac{\mathcal{Y}_{\mathbf{L}_{\boldsymbol{\mathcal{X}}}}}{\rho}\right\|_F^2 \right) \\ \min\limits_{C_{\boldsymbol{\mathcal{X}}}} \left(\lambda\mu\|\mathbf{C}_{\mathcal{X}}\|_1 + \frac{\rho}{2}\left\|\mathbf{C}_{\mathcal{X}} - \mathbf{C}_{\mathcal{X}_k} + \frac{\mathcal{Y}_{\mathbf{C}_k}}{\rho}\right\|_F^2 \right) \end{bmatrix}. \tag{29}$$

Problem (29) can be solved by updating $\mathbf{L}_{\mathcal{X}_{k+1}}$ and $\mathbf{C}_{\mathcal{X}_{k+1}}$ via singular value thresholding algorithm [54]. Let $\mathcal{S}_{\mathcal{T}} : \mathbb{R} \to \mathbb{R}$ denote the shrinkage operator $\mathcal{S}_{\mathcal{T}}[t] = \text{sgn}(t)\max(|t| - \mathcal{T}, 0)$. The updating $\mathbf{L}_{\mathcal{X}_{k+1}}$ and $\mathbf{C}_{\mathcal{X}_{k+1}}$ can be expressed as

$$\begin{bmatrix} \mathbf{L}_{\mathcal{X}_{k+1}} \\ \mathbf{C}_{\mathcal{X}_{k+1}} \end{bmatrix} = \begin{bmatrix} \boldsymbol{\mathcal{D}}_{\mu/\rho}\left(\mathbf{L}_{\mathcal{Z}_k} + \frac{\mathcal{Y}_{\mathbf{L}_{\boldsymbol{\mathcal{X}}}}}{\rho}\right) \\ \mathbf{C}_{\lambda\mu/\rho}\left(\mathbf{C}_{\mathcal{X}_k} + \frac{\mathcal{Y}_{\mathbf{C}_k}}{\rho}\right) \end{bmatrix} \tag{30}$$

where $\boldsymbol{\mathcal{D}}_{\mu/\rho}(\mathbf{L}_{\mathcal{Z}_k} + \mathcal{Y}_{\mathbf{L}_{\boldsymbol{\mathcal{X}}}}/\rho) = \boldsymbol{\mathcal{U}}\mathbf{C}_{\mu/\rho}(\mathbf{\Sigma})\boldsymbol{\mathcal{V}}^T$, $\boldsymbol{\mathcal{U}}$ and $\boldsymbol{\mathcal{V}}$ denote the left and right singular value vector of $\mathbf{L}_{\mathcal{Z}_k} + \mathcal{Y}_{\mathbf{L}_{\boldsymbol{\mathcal{X}}}}/\rho$, respectively, and $\mathbf{\Sigma}$ denotes the singular value matrix.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:23:47 UTC from IEEE Xplore. Restrictions apply.

TABLE I
ESP-LRSMD DETECTOR FOR SHIP DETECTION IN SLC SAR IMAGE

Algorithm: ESP-LRSMD

Input: $\mathbf{S_{st}} = [s_{st}^1, s_{st}^2, \ldots, s_{st}^{Na}]^T$;
User Parameters: $\lambda = \frac{1}{\sqrt{\max(N_f, N_t)}}$, $\mu = (\sqrt{N_f} + \sqrt{N_t})$;
Initialization: $L = 64$, $\mathcal{E} = 10^{-7}$, $\rho > 0$;
Repeat
  Obtain the Hankel matrix $\mathbf{T_n}$ of $s_{st}^n(\tau)$ by (24);
  $\mathbf{\Omega}_n = \mathbf{T}_n \mathbf{T}_n^H$;
  $[\mathbf{\Lambda}_n, \mathbf{E}_n] = \mathrm{EVD}(\mathbf{\Omega}_n)$;
  $\xi_n = \max[\mathbf{\Lambda}_n] = \max[\gamma_1^n, \gamma_2^n, \gamma_3^n, \ldots, \gamma_L^n]$;
  $n = n + 1$;
Until $n > N_a$
$\Phi(n) = [\xi_1, \xi_2, \cdots, \xi_{N_a}]$, update $\Phi(n)$ by (28);
Output: $\overline{\Phi}(n)$.
Repeat
If $\overline{\Phi}(n) = 0$, $\tilde{P}^n = \mathbf{0}$.
Else, $STFT[s_{st}^n(\tau)] = \mathbf{M}$,
  while not converged do
    Calculate $\mathbf{L}_{\mathcal{X}_{k+1}}$ and $\mathbf{C}_{\mathcal{X}_{k+1}}$ by (36);
    Calculate $\mathbf{L}_{\mathcal{Z}_{k+1}}$ and $\mathbf{C}_{\mathcal{Z}_{k+1}}$ by (38);
    Calculate $\boldsymbol{\mathcal{Y}}_{L_{k+1}}$ and $\boldsymbol{\mathcal{Y}}_{C_{k+1}}$ by (39);
  end while
  Output: $\mathbf{L}$.
  Calculate P^n by (41) and update P^n by (43);
 $n = n + 1$;
Until $n > N_a$.
Output: $\mathbf{P} = [\tilde{P}^1, \tilde{P}^2, \ldots, \tilde{P}^{N_a}]^T$.

Then, $\boldsymbol{\mathcal{Z}}_{\mathbf{k}}$ can be updated by fixing $\boldsymbol{\mathcal{X}}_{k+1}$ and $\boldsymbol{\mathcal{Y}}_{\mathbf{k}}$, which can be expressed as

$$\boldsymbol{\mathcal{Z}}_{k+1} = g(\boldsymbol{\mathcal{Z}}) + \langle \boldsymbol{\mathcal{Y}}_k, \boldsymbol{\mathcal{X}} \rangle + \frac{\rho}{2} \|\mathbf{M} - \boldsymbol{\mathcal{Z}}\|_F^2. \tag{31}$$

$\mathbf{L}_{\mathcal{Z}_{k+1}}$ and $\mathbf{C}_{\mathcal{Z}_{k+1}}$ can be updated as

$$\begin{bmatrix} \mathbf{L}_{\mathcal{Z}_{k+1}} \\ \mathbf{C}_{\mathcal{Z}_{k+1}} \end{bmatrix} = \begin{bmatrix} \frac{1}{1+\rho}\left(\boldsymbol{\mathcal{Y}}_{L_k} + \rho \mathbf{L}_{\mathcal{X}_{k+1}} + \mathbf{M} - \mathbf{C}_{\mathcal{Z}_k}\right) \\ \frac{1}{1+\rho}\left(\boldsymbol{\mathcal{Y}}_{C_k} + \rho \mathbf{C}_{\mathcal{X}_{k+1}} + \mathbf{M} - \mathbf{L}_{\mathcal{Z}_{k+1}}\right) \end{bmatrix}. \tag{32}$$

Then, $\boldsymbol{\mathcal{Y}}$ can be updated as

$$\begin{cases} \boldsymbol{\mathcal{Y}}_{L_{k+1}} = \boldsymbol{\mathcal{Y}}_{L_k} + \rho(\mathbf{L}_{\mathcal{X}_{k+1}} - \mathbf{L}_{\mathcal{Z}_{k+1}}) \\ \boldsymbol{\mathcal{Y}}_{C_{k+1}} = \boldsymbol{\mathcal{Y}}_{C_k} + \rho(\mathbf{C}_{\mathcal{X}_{k+1}} - \mathbf{C}_{\mathcal{Z}_{k+1}}). \end{cases} \tag{33}$$

The iteration termination condition is defined as

$$\frac{\left\| \left(\mathbf{L}_{\mathcal{X}_{k+1}}, \mathbf{C}_{\mathcal{X}_{k+1}}\right) - \left(\mathbf{L}_{\mathcal{X}_k}, \mathbf{C}_{\mathcal{X}_k}\right) \right\|_F}{\left\| \left(\mathbf{L}_{\mathcal{X}_k}, \mathbf{C}_{\mathcal{X}_k}\right) \right\|_F} < E. \tag{34}$$

Until meeting the iteration condition, the low-rank matrix $\mathbf{L} = \mathbf{L}_{\mathcal{X}_k}$ and the sparse matrix $\mathbf{C} = \mathbf{C}_{\mathcal{X}_k}$ can be output.

2) Ship Location and Sampling Point Extension: The low-rank matrix $\mathbf{L}$ constitutes the principal instantaneous frequency components of the ship targets, and the position of ship targets can be located by accumulating the energy in the frequency dimension

$$P^n(\tilde{\tau}) = \sum_{N_f} \mathbf{L}(N_f, N_t) \tag{35}$$

where $P^n(\tilde{\tau})$ is the position information of the ship targets. $\tilde{\tau}$ is the time sampling number after the processing of the time-limited window function, and its relationship with real-time sampling points can be expressed as

$$\tilde{\tau} = \left\lceil \frac{N_\tau}{\left| N_{\text{win}} - N_{\text{step}} \right|} \right\rceil \tag{36}$$

where N_τ, N_{win}, and N_{step} denote the total sampling points in time, length of time-limited window function, and step length, respectively. Then, employ the robust OTSU algorithm [55] to remove the false alarms in $P^n(\tilde{\tau})$, and it can be updated as

$$\tilde{P}^n(\tilde{\tau}) = \begin{cases} 0, & P^n(\tilde{\tau}) \geq \phi \\ 1, & P^n(\tilde{\tau}) < \phi. \end{cases} \tag{37}$$

where constants 1 and 0 label the location of the ship targets and false alarm, respectively, and ϕ is the threshold of segmentation obtained by the OTSU algorithm. As noted in (36), the sampling number after the sliding window operation of STFT is not equal to real-time sampling points in the time domain. Hence, the processing of range sampling point expansion, which is based on the processing of linear interpolation, is needed.

According to the detection result $\bar{\Phi}(n)$ in the first step and repeating the second step in a line-by-line manner, the ship position $\mathbf{P} = [\tilde{P}^1, \tilde{P}^2, \ldots, \tilde{P}^{N_a}]^T$ can be obtained. For clarity, the processing flow of the proposed detector is summarized in Table I.

V. NUMERICAL EXPERIMENTS

In this section, extensive experiments using spaceborne SAR SLC data from European Sentinel-1A and Chinese Gaofen-3 are performed to validate the effectiveness of the proposed detector. Two detectors, i.e., the bilateral CFAR detector [56] and the traditional RPCA-based [57] detector, are selected for comparison. In addition, three quality evaluation indices, namely, the precision rate (PR), recall rate (RR), and F1-score, are employed to verify the detection performance of the above detectors [58]. In this article, the SLC products acquired from the Sentinel-1A and Gaofen-3 are used for quantitative analysis experiments, and the ship label information is obtained from a high-resolution SAR images dataset (HRSID) [59].

A. Data Description

The spaceborne SAR datasets acquired from European Sentinel-1A are shown in Fig. 5, where the data acquisition areas are shown in Fig. 5(a) and (b). The scenes under normal sea-state conditions are shown in Fig. 5(c) and (e), the wind

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:23:47 UTC from IEEE Xplore. Restrictions apply.

TABLE II
EXPERIMENTAL DATA INFORMATION

SAR Image	Fig. 5 (c)	Fig. 5 (d)	Fig. 5 (e)	Fig. 5 (f)	Fig. 6 (a)	Fig. 6 (b)
Sensor	Sentinel-1A	Sentinel-1A	Sentinel-1A	Sentinel-1A	Gaofen-3	Gaofen-3
Band	C	C	C	C	C	C
Mode	SM	SM	SM	SM	SL	SL
Polarization	VV	VV	VV	VV	HH	HH
Acquisition Date	25 Aug. 2019	10 Mar. 2019	15 July 2019	31 Oct. 2019	14 Feb. 2021	21 Oct. 2020
Acquired region	Port Sao Paulo, Brazil	Port Sao Paulo, Brazil	Port Houston, America	Port Houston, America	Esperance, Australia	Esperance, Australia
Location	W 46.2, S 24.0	W 46.2, S 24.0	W 94.4, N 29.1	W 94.4, N 29.1	E 127.0, N 31.2	E 127.0, N 31.2
Resolution (range×azimth)	5 m × 5 m	5 m × 5 m	5 m × 5 m	5 m × 5 m	1 m × 1 m	1 m × 1 m
Size (range×azimth)	7100 × 8000	7500 × 8000	8000 × 5118	8000 × 5118	13600 × 30970	13600 × 30970
Product level	SLC	SLC	SLC	SLC	SLC	SLC

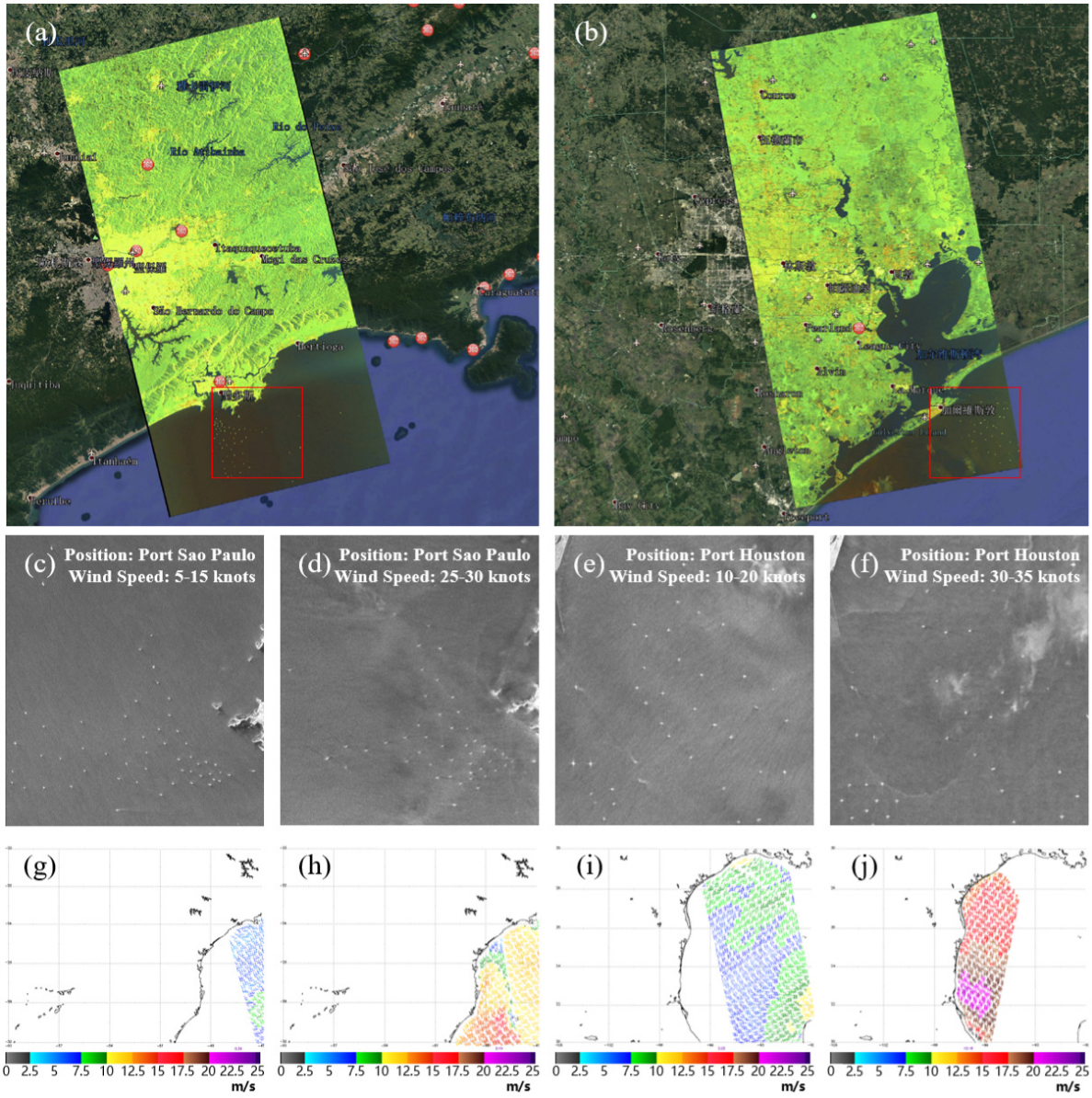


Fig. 5. Spaceborne SAR datasets acquired from European Sentinel-1A. (a) Nearshore scene in the Port of Sao Paulo, Brazil, acquired by Sentinel-1A. (b) Nearshore scene in the Port of Houston, USA, acquired by Sentinel-1A. (c)–(f) ROIs in (a) and (b). (g)–(j) Wind speed of the experimental data in (c)–(f).

speed is moderate, and scenes under high sea-state conditions are shown in Fig. 5(d) and (f). The wind field vector diagrams of the experimental data are given in Fig. 5(g)–(j). Another spaceborne SAR dataset acquired from Chinese Gaofen-3 is

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:23:47 UTC from IEEE Xplore. Restrictions apply.

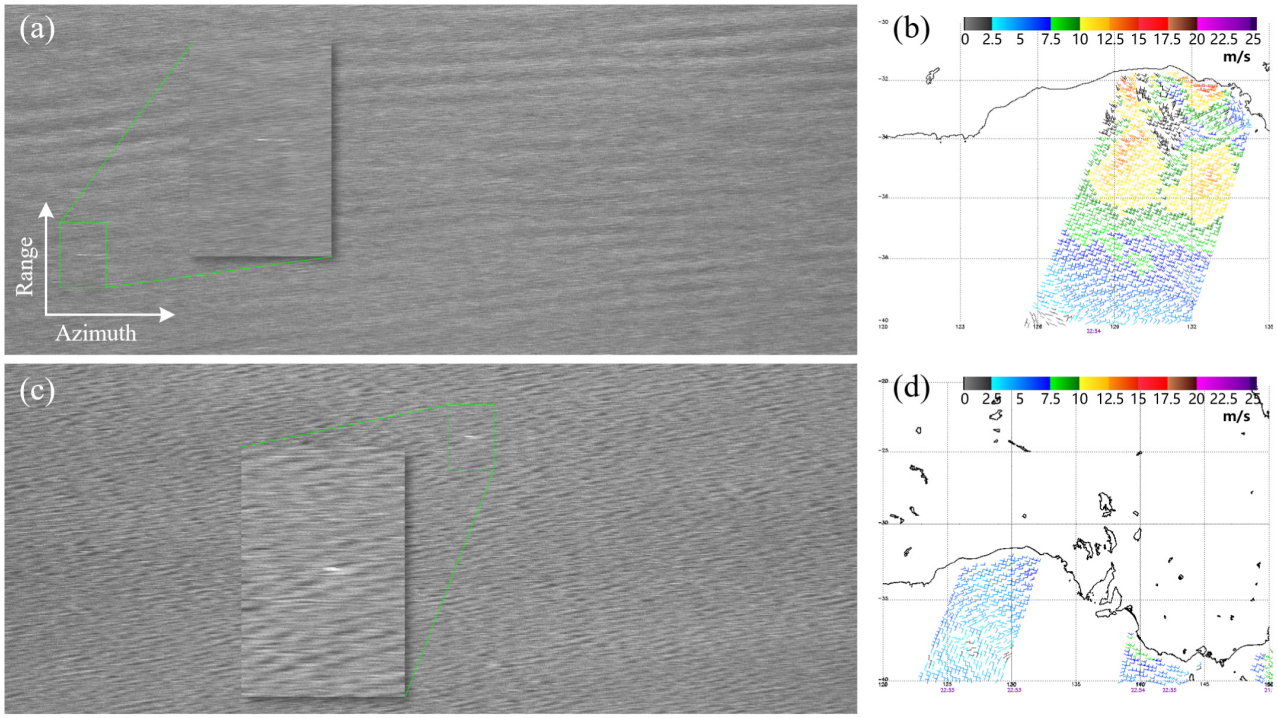


Fig. 6. Spaceborne SAR datasets acquired from Chinese Gaofen-3. (a) and (c) Far shore scene in Esperance, Australia, acquired by Gaofen-3. (b) and (d) Wind speed of the experimental data in (a) and (c), respectively.

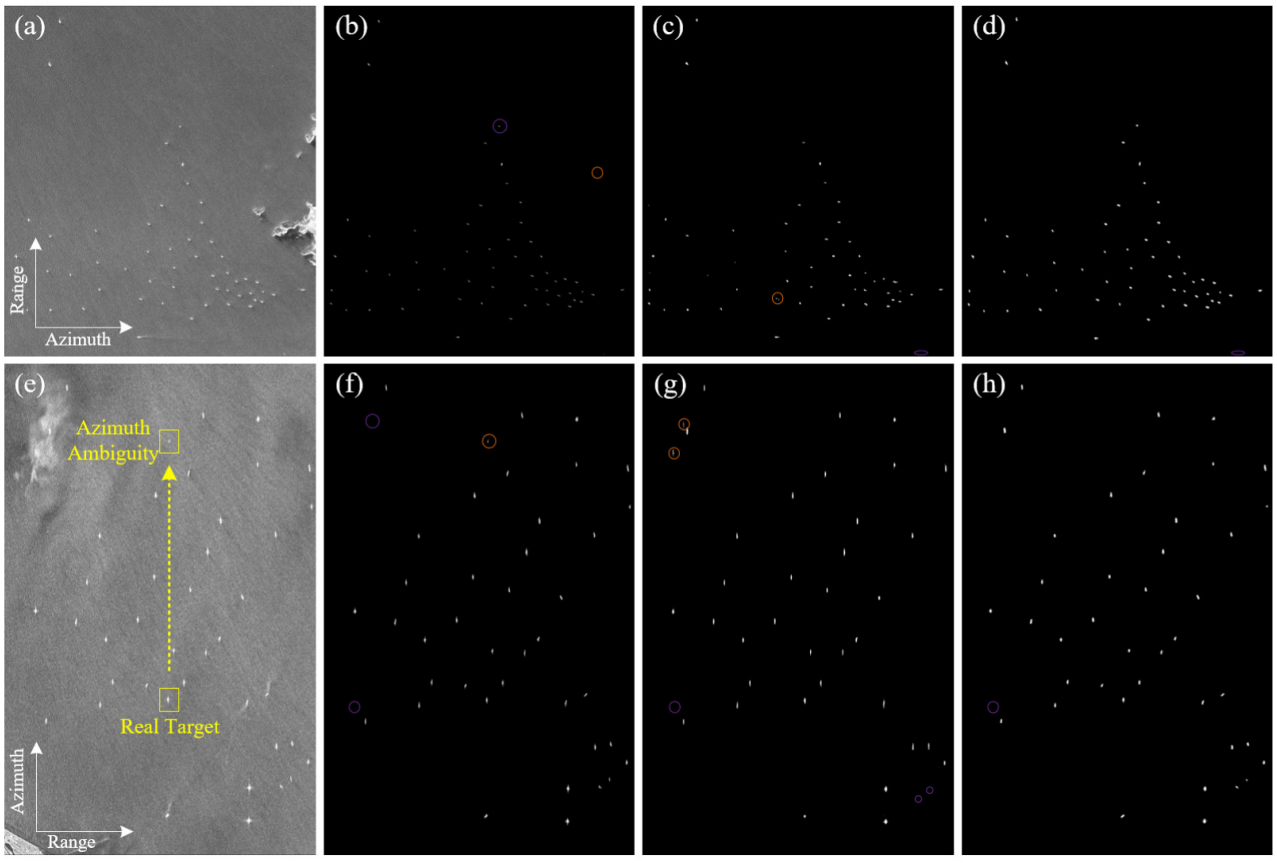


Fig. 7. Detection results under normal sea-state conditions. (a) and (e) Amplitude image of experimental scene. (b) and (f) Detection results for the bilateral CFAR detector (c) and (g) Detection results for the traditional RPCA-based detector. (d) and (h) Detection results for the proposed detector.

shown in Fig. 6, where the data acquisition area is located in Esperance, Australia. The wind field vector diagrams (acquired in the Advanced Scatterometer ASCAT METOP-A, available at https://manati.star.nesdis.noaa.gov) of the experimental data

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:23:47 UTC from IEEE Xplore. Restrictions apply.

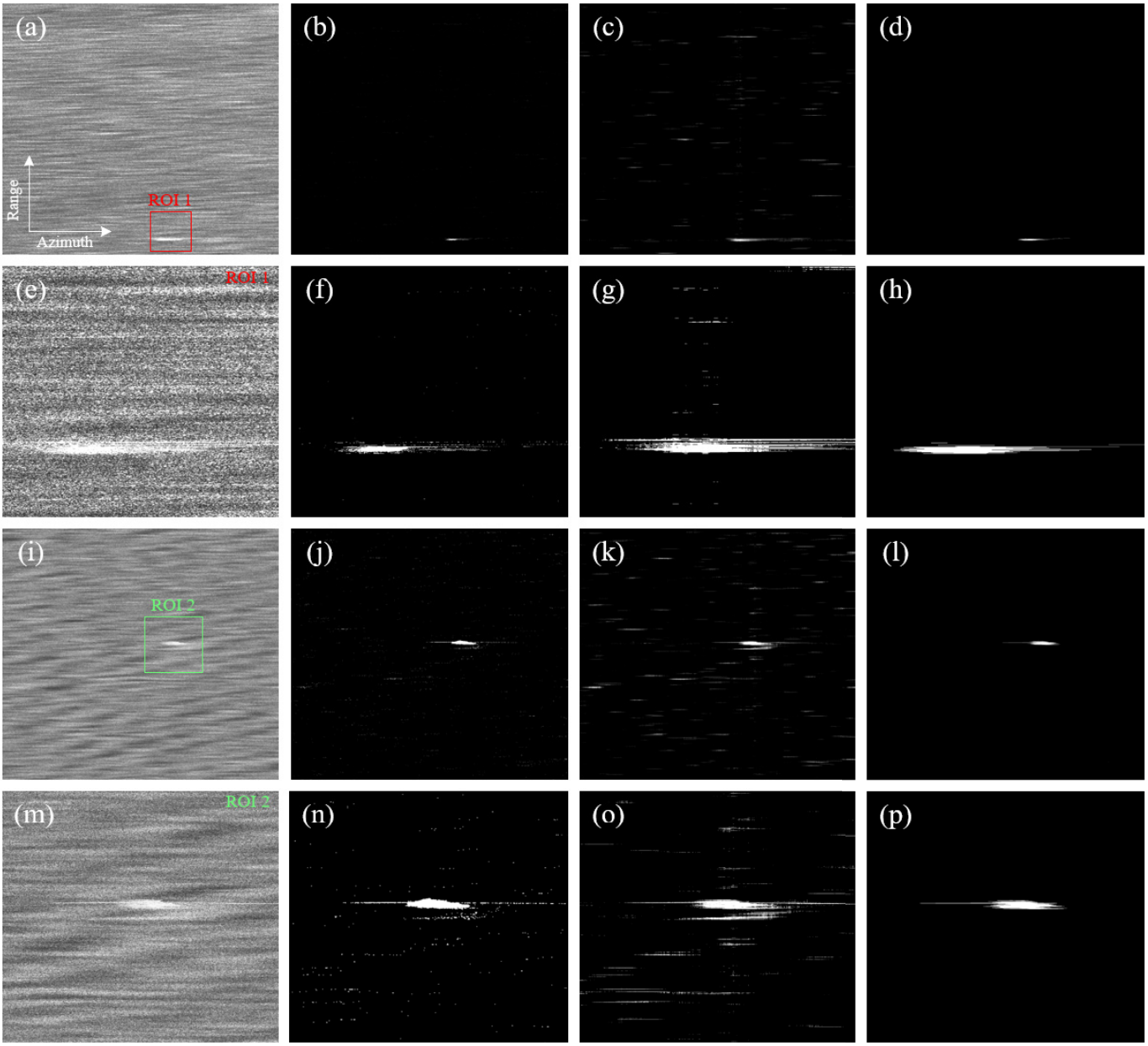


Fig. 8. Detection results for far shore scene under severe sea state. (a) and (i) Amplitude image of experimental scene. (b) and (j) Detection results for the bilateral CFAR detector (c) and (k) Detection results for the traditional RPCA-based detector. (d) and (l) Detection results for the proposed detector. (e)–(h) First ROI in (a)–(d), respectively. (m)–(p) Second ROI in (i)–(l), respectively.

are given in Fig. 6(b) and (d). The detailed information on the experimental data is listed in Table II.

B. Experimental Results and Analysis

1) On the Detection Performance Under Normal Sea State: Fig. 7 shows the ship detection results for single-pol SAR images acquired from Sentinel-1A, whose working mode is SM. The scenes of data acquisition are located in the Port of Sao Paulo, Brazil, and the Port of Houston, USA. Fig. 7(a) and (e) shows the experimental data, and Fig. 7(b)–(d), (f), and (g) shows the detection results for the bilateral CFAR detector, the traditional RPCA-based detector, and the proposed detector, respectively. The orange circles mark the ships that are incorrectly detected, whereas the purple circles mark the ones that are misdetected.

As shown in Fig. 7(a), wind speed is moderate, and the detection performance of the three detectors is well since the scene is homogeneous and the surface is flat. The yellow rectangle in Fig. 7(e) marks the azimuth ambiguity. The azimuth ambiguity is the interference that exists in the SAR system, which presents as the ghost of the strong target in the azimuth [60]. As shown in Fig. 7(f), the bilateral CFAR detector mistakenly classified the azimuth ambiguity as ship targets. This phenomenon is because the bilateral CFAR detector is performed on SAR amplitude imagery and the strong targets' ambiguity is similar to itself in amplitude. The proposed detector employs the low-rank feature of the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:23:47 UTC from IEEE Xplore. Restrictions apply.

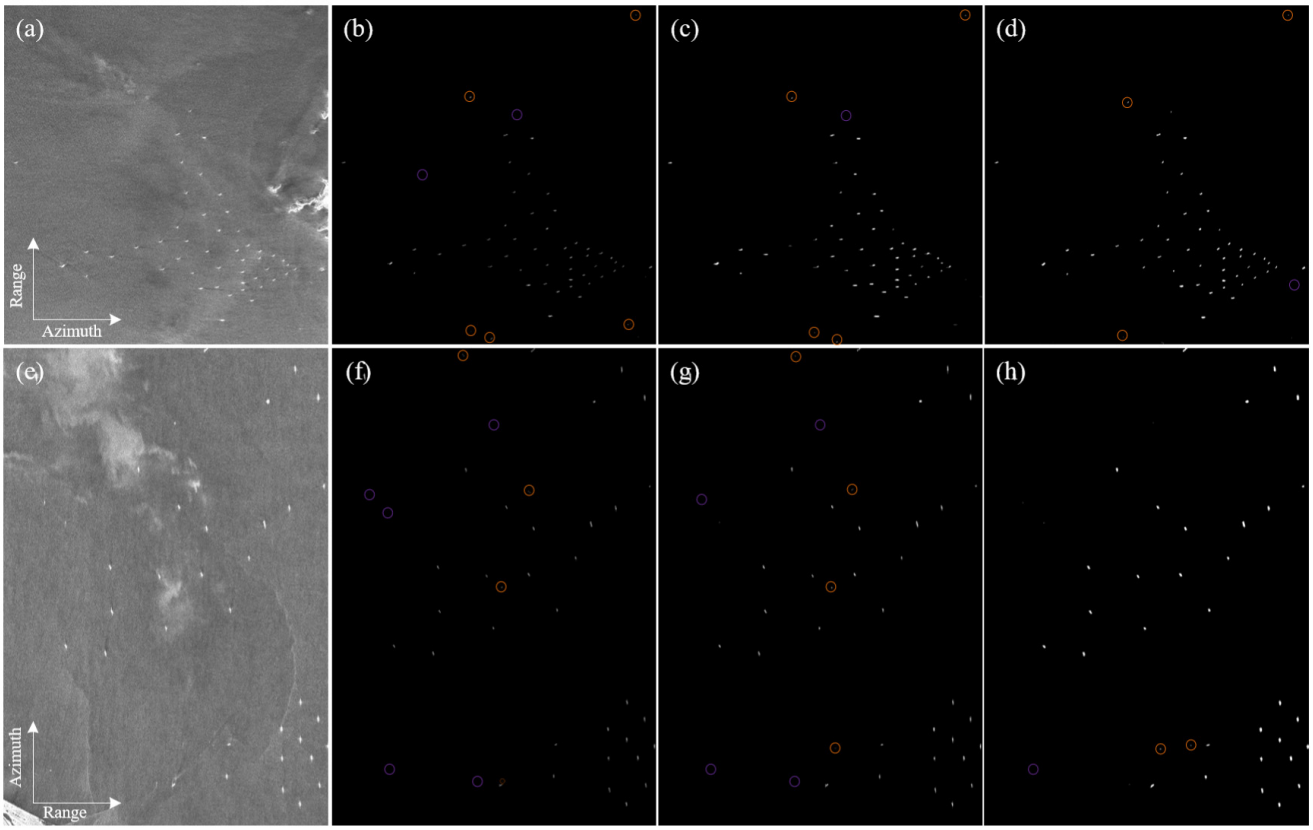


Fig. 9. Detection results under high sea-state conditions. (a) and (e) Amplitude image of experimental scene. (b) and (f) Detection results for the bilateral CFAR detector (c) and (g) Detection results for the traditional RPCA-based detector. (d) and (h) Detection results for the proposed detector.

instantaneous frequency of the ship targets, and therefore, it can eliminate the interference of the azimuth ambiguity by LRMSD technology. The upper left section in Fig. 7(e) marks the sea wave. Although the scattering intensity of the wave is higher than that of the sea clutter, its energy distribution is sparse, resulting in the maximum eigenvalue in the characteristic subspace domain being significantly lower than that of the focused ship target. As shown in Fig. 7(g), the traditional RPCA-based detector mistakenly classified the wave as ship targets, whereas the proposed detector does not.

2) On the Detection Performance Under Severe Sea State: Two scenes of Gaofen-3 SLC data are employed to qualitatively analyze the detection performance of the proposed detector under the severe sea state. The experimental results are shown in Fig. 8, where two ROIs mark the hard to distinguished ships. Influenced by the ocean wind, the sea surface in this scene is inhomogeneous and the ship shakes violently, which is difficult to focus on a ship target. In Fig. 8, the first column is the experimental image and its ROIs, the second column is the detection result of the bilateral CFAR detector, the third column is the detection result of the RPCA-based detector, and the fourth column is the detection result of the proposed detector.

As shown in Fig. 8(f) and (n), the bilateral CFAR detector successfully detects the ship under a severe sea state, but meanwhile, it mistakenly detects some false alarms. As shown in Fig. 8(f) and (n), the same results were produced on the RPCA-based detector. It is worth noting that in the experimental results, the contours of the detected ships are slightly different from the original one, and a large number of unnecessary false alarms are detected. The causes of these phenomena are given as follows. First, the RPCA-based detector holds that the ships are sparsely distributed on the sea surface, while the sea clutters can be considered as low-rank structures, and based on the low-rank matrix decomposition, the sparse ship targets can be extracted. The viewpoint is valid in the far shore scene under a normal sea state, but it is not the case under severe sea state because the inhomogeneous sea surface has strong scattering properties. In this case, the ships are still sparse, whereas the sea clutters are no longer low-rank. Second, the scene under severe sea state can be regarded as sparse ships mixed with high-rank sea clutter, and therefore, the "low-rank + sparse" model-based RPCA technology is no longer suitable for extracting ship targets since it would overpenalize the singular value causing low-rank approximation failure. The theoretical basis of the effectiveness of the proposed detector is different from the traditional RPCA-based detector. In the first step of the proposal, some false alarms are eliminated, and in the second step, by means of the matrix decomposition technology, the ships with low-rank features can be extracted in the time–frequency domain. As shown in Fig. 8(h) and (p), the performance of the proposed detector is better than other detectors.

Furthermore, two scenes of Sentinel-1A SLC data are employed to quantitatively analyze the detection performance of the proposed detector under high sea-state conditions,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:23:47 UTC from IEEE Xplore. Restrictions apply.

TABLE III
QUANTITATIVE EVALUATION INDICES COMPARISON WITH THREE SAR SHIP DETECTOR

Scene	Sea-State	Detector	PR (%)	RR (%)	F1-score (%)	Time(s)
Port Sao Paulo 7100×8000	Normal (wind speed 5-20 knots)	Bilateral CFAR	98.21	98.21	98.21	72.94
		RPCA	96.49	98.21	96.49	218.54
		Proposed	100.00	98.21	100.00	79.85
Port Houston 8000×5118		Bilateral CFAR	97.43	92.68	94.99	37.63
		RPCA	94.87	92.50	93.67	112.77
		Proposed	100.00	97.50	100.00	42.61
Port Sao Paulo 7500×8000	High (wind speed 5-20 knots)	Bilateral CFAR	89.79	95.65	92.63	81.37
		RPCA	91.84	97.83	94.74	244.23
		Proposed	93.88	97.87	95.83	81.58
Port Houston 8000×5118		Bilateral CFAR	89.47	87.50	88.47	37.92
		RPCA	89.47	89.74	89.74	112.77
		Proposed	94.74	97.29	95.99	41.97

which is shown in Fig. 9. Compared with the experimental data in Fig. 6, the sea surface is turbulent and some ships are indistinguishable. The detection performance of the three methods is affected to different degrees, where the marked circles indicating incorrectly and misdetected show that the performance of the proposed detector under the high sea-state conditions is better than other detectors.

Three quality evaluation indices, i.e., PR, RR, and F1-score, are calculated and summarized in Table III. The ship label information is obtained from HRSID. The PR is defined as the ratio of the ships that is detected correctly in all detected ships, which can be expressed as

$$\mathrm{PR}(\mathrm{TP},\mathrm{FP}) = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \tag{38}$$

where TP denotes the true positives, i.e., the numbers of the ships that are correctly detected, and FP denotes the false positive, i.e., the number of ships incorrectly detected (false alarm).

The RR is a measure of coverage, and it is defined as the ratio of the ships being detected in all ground truths, which can be expressed as

$$\mathrm{RR}(\mathrm{TP},\mathrm{FN}) = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \tag{39}$$

where FN denotes the false negatives and it can be understood as the missed detection.

The F1-score is defined as a harmonic mean between PR and RR, which can be expressed as

$$\mathrm{F1}-\mathrm{score}(\mathrm{PR},\mathrm{RR}) = 2\times\frac{\mathrm{PR}\times\mathrm{RR}}{\mathrm{PR}+\mathrm{RR}}. \tag{40}$$

It can be seen from Table III that the PR of the proposed detector is better than other methods, which means that the proposed method is not sensitive to the false alarm. The RR of the PRCA-based detector is better than other methods, whereas its RR is as unsatisfactory as that of the bilateral CFAR detector, which causes it to reduce the level of F1-score itself. According to F1-score reflected, it can be seen that the comprehensive detection performance of the proposed method is better than other methods, which verifies its superiority and effectiveness. All the programs are tested with MATLAB R2018a on a computer with a 3.85-GHz CPU AMD Ryzen 5 2600 and 32-GB RAM. From Table III, it can be found that the running time of the proposed detector and the bilateral CFAR detector is significantly faster than that of the traditional RPCA detector.

From the qualitative and quantitative analysis of experimental results, it can be concluded that the detection performance of the bilateral CFAR detector is average, whereas the RPCA-based detector is better than the bilateral CFAR detector under normal sea state. The detection performance of the proposed detector is better than that of the other detectors.

VI. CONCLUSION

In this article, aiming at ship detection in SAR SLC products, a two-step detector combining ESP and LRSMD technology is proposed. The potential of complex-valued SAR images in the field of SAR ship detection is mined. Specifically, from the theoretical foundation and experimental analysis, the strong scattering feature of ship targets in the eigen-subspace is verified, and the low-rank feature of the ship targets in the time–frequency domain is demonstrated. The proposal is different from the traditional model-driven detectors in which it detects the occurrence of ship targets in the first step and, on this basis, separates the low-rank instantaneous frequency in the second step to locate the ship, which can effectively reduce false alarms. The precondition for the effectiveness of the proposed detector is that the ship and sea clutter have low-rank and sparsity features in the time–frequency domain.

Experiments based on European Sentinel-1A and Chinese Gaofen-3 single-pol SAR SLC images are performed. Specifically, experimental results via SAR SLC images under normal sea state show that the F1-score of ship detection via the proposed detector reaches an acceptable level, which can prove that a better detection performance of the proposed detector can be achieved. Experimental results via SAR SLC images under high sea-state show that the proposed detector is effective for detection tasks under complex sea states.

ACKNOWLEDGMENT

The authors would like to thank the National Satellite Ocean Application Service (NSOAS) for providing the GaoFen-3 SAR data, the European Space Agency (ESA) for providing

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:23:47 UTC from IEEE Xplore. Restrictions apply.

the Sentinel-1A SAR data, and the National Oceanic and the Atmospheric Administration (NOAA) and the National Environmental Satellite Data and Information Service (NESDIS) for providing the ocean surface winds product.

REFERENCES

[1] K. Eldhuset, "An automatic ship and ship wake detection system for spaceborne SAR images in coastal regions," *IEEE Trans. Geosci. Remote Sens.*, vol. 34, no. 4, pp. 1010–1019, Jul. 1996.

[2] A. Moreira, P. Prats-Iraola, M. Younis, G. Krieger, I. Hajnsek, and K. P. Papathanassiou, "A tutorial on synthetic aperture radar," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 1, pp. 6–43, Apr. 2013.

[3] A. Reigber *et al.*, "Very-high-resolution airborne synthetic aperture radar imaging: Signal processing and applications," *Proc. IEEE*, vol. 101, no. 3, pp. 759–783, Mar. 2013.

[4] D. Tao, S. N. Anfinsen, and C. Brekke, "Robust CFAR detector based on truncated statistics in multiple-target situations," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 117–134, Jan. 2016.

[5] T. Zhang, W. Wang, Z. Yang, J. Yin, and J. Yang, "Ship detection from PolSAR imagery using the hybrid polarimetric covariance matrix," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 9, pp. 1575–1579, Sep. 2021.

[6] T. Zhang, Z. Yang, and H. Xiong, "PolSAR ship detection based on the polarimetric covariance difference matrix," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 7, pp. 3348–3359, Jul. 2017.

[7] K. Fu, J. Fu, Z. Wang, and X. Sun, "Scattering-keypoint-guided network for oriented ship detection in high-resolution and large-scale SAR images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 11162–11178, 2021.

[8] D. J. Crisp, "The state-of-the-art in ship detection in synthetic aperture radar imagery," *Organic Lett.*, vol. 35, no. 42, pp. 2165–2168, 2004.

[9] G. B. Goldstein, "False-alarm regulation in log-normal and Weibull clutter," *IEEE Trans. Aerosp. Electron. Syst.*, vol. AES-9, no. 1, pp. 84–92, Jan. 1973.

[10] S. Kuttikkad and R. Chellappa, "Non-Gaussian CFAR techniques for target detection in high resolution SAR images," in *Proc. 1st Int. Conf. Image Process.*, Nov. 1994, pp. 910–914.

[11] M. Migliaccio, L. Huang, and A. Buono, "SAR speckle dependence on ocean surface wind field," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5447–5455, Aug. 2019.

[12] A. Balleri, A. Nehorai, and J. Wang, "Maximum likelihood estimation for compound-Gaussian clutter with inverse gamma texture," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 43, no. 2, pp. 775–779, Apr. 2007.

[13] X. Qin, S. Zhou, H. Zou, and G. Gao, "A CFAR detection algorithm for generalized gamma distributed background in high-resolution SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 10, no. 4, pp. 806–810, Jul. 2013.

[14] D. Tao, S. N. Anfinsen, and C. Brekke, "Robust CFAR detector based on truncated statistics in multiple-target situations," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 117–134, Jan. 2016.

[15] D.-S. Han, "Detection performance of CFAR detectors based on order statistics for partially correlated chi-square targets," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 36, no. 4, pp. 1423–1429, Oct. 2000.

[16] J. Ai, X. Qi, W. Yu, Y. Deng, F. Liu, and L. Shi, "A new CFAR ship detection algorithm based on 2-D joint log-normal distribution in SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 7, no. 4, pp. 806–810, Oct. 2010.

[17] J. Ai, X. Yang, J. Song, Z. Dong, L. Jia, and F. Zhou, "An adaptively truncated clutter-statistics-based two-parameter CFAR detector in SAR imagery," *IEEE J. Ocean. Eng.*, vol. 43, no. 1, pp. 267–279, Jan. 2018.

[18] M. Migliaccio, G. Ferrara, A. Gambardella, F. Nunziata, and A. Sorrentino, "A physically consistent speckle model for marine SLC SAR images," *IEEE J. Ocean. Eng.*, vol. 32, no. 4, pp. 839–847, Oct. 2007.

[19] G. Ferrara, M. Migliaccio, F. Nunziata, and A. Sorrentino, "Generalized-K (GK)-based observation of metallic objects at sea in full-resolution synthetic aperture radar (SAR) data: A multipolarization study," *IEEE J. Ocean. Eng.*, vol. 36, no. 2, pp. 195–204, Apr. 2011.

[20] M. Weiss, "Analysis of some modified cell-averaging CFAR processors in multiple-target situations," *IEEE Trans. Aerosp. Electron. Syst.*, vol. AES-18, no. 1, pp. 102–114, Jan. 1982.

[21] V. G. Hansen, "Constant false alarm rate processing in search radars," in *Proc. IEE Conf. Publication 105, Radar-Present Future*, 1973, pp. 325–332.

[22] J. Ai, R. Tian, Q. Luo, J. Jin, and B. Tang, "Multi-scale rotation-invariant Haar-like feature integrated CNN-based ship detection algorithm of multiple-target environment in SAR imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 12, pp. 10070–10087, Dec. 2019.

[23] A. Gambardella, F. Nunziata, and M. Migliaccio, "A physical full-resolution SAR ship detection filter," *IEEE Geosci. Remote Sens. Lett.*, vol. 5, no. 4, pp. 760–763, Oct. 2008.

[24] M. Migliaccio, F. Nunziata, A. Montuori, and R. L. Paes, "Single-look complex COSMO-SkyMed SAR data to observe metallic targets at sea," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 5, no. 3, pp. 893–901, Jun. 2012.

[25] X. X. Zhu *et al.*, "Deep learning meets SAR: Concepts, models, pitfalls, and perspectives," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 4, pp. 143–172, Dec. 2021.

[26] J. Li, C. Qu, and J. Shao, "Ship detection in SAR images based on an improved faster R-CNN," in *Proc. SAR Big Data Era, Models, Methods Appl. (BIGSARDATA)*, Nov. 2017, pp. 1–6.

[27] K. Fu, J. Fu, Z. Wang, and X. Sun, "Scattering-keypoint-guided network for oriented ship detection in high-resolution and large-scale SAR images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 11162–11178, 2021.

[28] J. Li, S. Gou, R. Li, J.-W. Chen, and X. Sun, "Ship segmentation via encoder–decoder network with global attention in high-resolution SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[29] J. Li *et al.*, "A ship detection method based on cascade CNN in SAR images," *Control Decis.*, vol. 34, no. 10, pp. 2191–2197, 2019.

[30] Z. Lin, K. Ji, X. Leng, and G. Kuang, "Squeeze and excitation rank faster R-CNN for ship detection in SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 5, pp. 751–755, May 2019.

[31] Z. Cui, Q. Li, Z. Cao, and N. Liu, "Dense attention pyramid networks for multi-scale ship detection in SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 8983–8997, Nov. 2019.

[32] X. Sun, P. Wang, C. Wang, Y. Liu, and K. Fu, "PBNet: Part-based convolutional neural network for complex composite object detection in remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 173, pp. 50–65, Mar. 2021.

[33] J. Redmon, S. Divvala, R. B. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. Comput. Vis. Pattern Recognit.* Las Vegas, NV, USA, Jun. 2016, pp. 779–788.

[34] Z. Wang, L. Du, J. Mao, B. Liu, and D. Yang, "SAR target detection based on SSD with data augmentation and transfer learning," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 1, pp. 150–154, Jan. 2019.

[35] Y. Zhao, L. Zhao, B. Xiong, and G. Kuang, "Attention receptive pyramid network for ship detection in SAR images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 2738–2756, 2020.

[36] T. Zhang and X. Zhang, "High-speed ship detection in SAR images based on a grid convolutional neural network," *Remote Sens.*, vol. 11, no. 10, p. 1206, May 2019.

[37] J. Tang, J. Cheng, D. Xiang, and C. Hu, "Large-difference-scale target detection using a revised Bhattacharyya distance in SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[38] T. Li, Z. Liu, L. Ran, and R. Xie, "Target detection by exploiting superpixel-level statistical dissimilarity for SAR imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 4, pp. 562–566, Apr. 2018.

[39] J. Sánchez, F. Perronnin, T. Mensink, and J. Verbeek, "Image classification with the Fisher vector: Theory and practice," *Int. J. Comput. Vis.*, vol. 105, no. 3, pp. 222–245, 2013.

[40] H. Lin, H. Chen, K. Jin, L. Zeng, and J. Yang, "Ship detection with superpixel-level Fisher vector in high-resolution SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 247–251, Feb. 2020.

[41] X. Wang, G. Li, X. Zhang, and Y. He, "Ship detection in SAR images via local contrast of Fisher vectors," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 9, pp. 6467–6479, Sep. 2020.

[42] L. Han, D. Liu, and D. Guan, "Ship detection in SAR images by saliency analysis of multiscale superpixels," *Remote Sens. Lett.*, vol. 13, no. 7, pp. 708–715, Jul. 2022.

[43] L. Zhang, Z. Zhang, S. Lu, D. Xiang, and Y. Su, "Fast superpixel-based non-window CFAR ship detector for SAR imagery," *Remote Sens.*, vol. 14, no. 9, p. 2092, Apr. 2022.

[44] A. Marino, M. J. Sanjuan-Ferrer, I. Hajnsek, and K. Ouchi, "Ship detection with spectral analysis of synthetic aperture radar: A comparison of new and well-known algorithms," *Remote Sens.*, vol. 7, no. 5, pp. 5416–5439, 2015.

[45] W. Wu, X. Li, H. Guo, L. Ferro-Famil, and L. Zhang, "Noncircularity parameters and their potential applications in UHR MMW SAR data sets," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 10, pp. 1547–1551, Oct. 2016.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:23:47 UTC from IEEE Xplore. Restrictions apply.

[46] K. El-Darymli, P. Mcguire, E. W. Gill, D. Power, and C. Moloney, "Characterization and statistical modeling of phase in single-channel synthetic aperture radar imagery," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 51, no. 3, pp. 2071–2092, Jul. 2015.
[47] X. Leng, K. Ji, S. Zhou, and X. Xing, "Ship detection based on complex signal kurtosis in single-channel SAR imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6447–6461, Sep. 2019.
[48] B. D. Rao and K. V. S. Hari, "Weighted subspace methods and spatial smoothing: Analysis and comparison," *IEEE Trans. Signal Process.*, vol. 41, no. 2, pp. 788–803, Feb. 1993.
[49] Y. Zhang and B. P. Ng, "MUSIC-like DOA estimation without estimating the number of sources," *IEEE Trans. Signal Process.*, vol. 58, no. 3, pp. 1668–1676, Mar. 2010.
[50] X. Zhuang, X. Cui, M. Lu, and Z. Feng, "Low-complexity method for DOA estimation based on ESPRIT," *J. Syst. Eng. Electron.*, vol. 21, no. 5, pp. 729–733, Oct. 2010.
[51] D. Gabor, "Theory of communication," *J. IEE*, vol. 93, no. 3, pp. 429–457, Nov. 1946.
[52] E. J. Candès and B. Recht, "Exact matrix completion via convex optimization," *Found. Comput. Math.*, vol. 9, no. 6, pp. 717–772, 2009.
[53] J. Wright, A. Ganesh, S. Rao, Y. Peng, and Y. Ma, "Robust principal component analysis: Exact recovery of corrupted low-rank matrices via convex optimization," in *Proc. 22nd Int. Conf. Neural Inf. Process. Syst. (NIPS)*. Red Hook, NY, USA, 2009, pp. 2080–2088.
[54] J. A. Hartigan and M. A. Wong, "Algorithm AS 136: A k-means clustering algorithm," *Appl. Statist.*, vol. 28, no. 1, pp. 100–108, Jan. 1979.
[55] J. F. Cai, E. J. Candès, and Z. Shen, "A singular value thresholding algorithm for matrix completion," *SIAM J. Optim.*, vol. 20, no. 4, pp. 1956–1982, Feb. 2010.
[56] N. Otsu, "A threshold selection method from gray-level histograms," *IEEE Trans. Syst., Man, Cybern.*, vol. SMC-9, no. 1, pp. 62–66, Jan. 1979.
[57] X. Leng, K. Ji, K. Yang, and H. Zou, "A bilateral CFAR algorithm for ship detection in SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 7, pp. 1536–1540, Jul. 2015.
[58] S. Song and J. Yang, "Ship detection in SAR images by robust principal component analysis," *J. Tsinghua Univ. Sci. Technol.*, vol. 55, no. 8, pp. 844–848, Aug. 2015.
[59] D. Li, Q. Liang, H. Liu, Q. Liu, H. Liu, and G. Liao, "A novel multidimensional domain deep learning network for SAR ship detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022.
[60] S. Wei *et al.*, "HRSID: A high-resolution SAR images dataset for ship detection and instance segmentation," *IEEE Access*, vol. 8, pp. 120234–120254, 2020.
[61] J. Ai, G. Fan, Y. Mao, J. Jin, M. Xing, and H. Yan, "An improved SRGAN based ambiguity suppression algorithm for SAR ship target contrast enhancement," *IEEE Trans. Geosci. Remote Sens.*, vol. 19, 2021, Art. no. 4018705.



Zongsen Lv (Graduate Student Member, IEEE) was born in Henan, China, in 1996. He received the B.S. degree in computer science and technology from Henan University, Kaifeng, China, in 2018, where he is currently pursuing the M.S. degree with the School of Computer and Information Engineering.

His research interests include synthetic aperture radar (SAR) imaging, signal processing, and radio frequency interference (RFI) mitigation.



Jing Lu received the B.S. degree in business administration from Nankai University, Tianjin, China, in 2008, and the M.S. degree in master of business administration (MBA) from Beijing Normal University, Beijing, China, in 2013.

Since June 2014, she has been working at the Land Satellite Remote Sensing Application Center, Ministry of Natural Resources, Beijing. She is engaged in the management of remote sensing innovation platform and the development of remote sensing application technology.



Qing Wang received the B.S. degree in remote sensing science and technology from Chang'an University, Xi'an, China, in 2007, the M.S. degree in microwave remote sensing from the University of Chinese Academy of Sciences, Beijing, China, in 2010, and the Ph.D. degree in photogrammetry and remote sensing from Peking University, Beijing, in 2014.

His research interests include computer vision and synthetic aperture radar image processing.



Zhengwei Guo (Member, IEEE) received the B.S. degree in radio technology engineering from the Huazhong University of Science and Technology, Wuhan, China, in 1984.

After receiving her B.S. degree, she joined Henan University, Kaifeng, China, as an Assistant Professor. Since 2008, she has been a Professor with the School of Computer and Information Engineering, Henan University. She has published several papers including peer-reviewed journals, conference proceedings, patents, and software copyrights. Her research focuses on management system software, data processing system, and information security technology. Her projects on synthetic aperture radar (SAR) image applications and information processing system have been supported by the National Natural Science Foundation of China and the Department of Science and Technology of Henan Province.

Ms. Guo received the First-Prize Scientific and Technological Progress Award of Henan Province.



Ning Li (Member, IEEE) received the B.S. degree in electronics information engineering from Northeast Forestry University, Harbin, China, in 2009, the M.S. degree in communication and information system from the Nanjing University of Aeronautics and Astronautics, Nanjing, China, in 2012, and the Ph.D. degree from the Institute of Electronics, Chinese Academy of Sciences (IECAS), Beijing, China, in 2015.

In July 2015, he joined the Department of Space Microwave Remote Sensing System, IECAS, where he was an Assistant Professor. Since December 2017, he has been a Full Professor with the School of Computer and Information Engineering, Henan University, Kaifeng, China. His research interests include synthetic aperture radar (SAR) and inverse SAR signal and image processing.

Dr. Li was a recipient of the Special Prize of President Scholarship for Postgraduate Students from the University of Chinese Academy of Sciences in 2015.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:23:47 UTC from IEEE Xplore. Restrictions apply.